

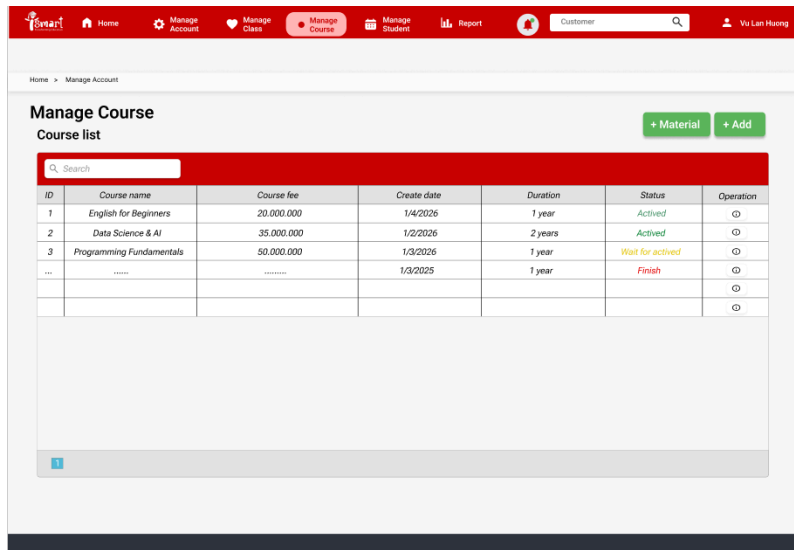
Admin

1. USER STORY: Admin View Courses

Card	- As an Admin - I want to view and search courses, including viewing course details - So that I can monitor and access full course information
Summary	This user story allows Admin Admin can view course list, search courses, and view detailed course information, the course status to help Admin distinguish between active and draft courses.
In-Scope:	- View the course list, search by keyword, and view specific course details.
Out of scope	- Modify course data.
Trigger	Admin clicks the [Manage Course] menu on the Sidebar.
Pre-condition	- Admin must be logged into the system - Admin role must have permission to manage courses - System database must be available
Business rule and error message	- System must retrieve course data from database Error: <i>Unable to load course data"</i> - If no courses exist in the system Error: No courses available in the system. - Duration must be greater than 0 Error: <i>Duration must be greater than 0.</i> - Fee must be a valid positive number Error: <i>Course fee must be greater than 0.</i> - Course cannot be deleted if there are classes associated with the course Error: <i>Course cannot be deleted because classes are associated with it.</i>

- created_by must reference the logged-in Admin (User.user_id)

Screen design



Screen description

Course Management screen allows Admin to view and search course information.

Includes:

- Search bar (search by course name)
- Course table:
 - Course Name
 - Duration
 - Fee
 - Status
 - Created At
- Click row / View button → open course detail
- Add New button
- (i) to Edit button and edit button

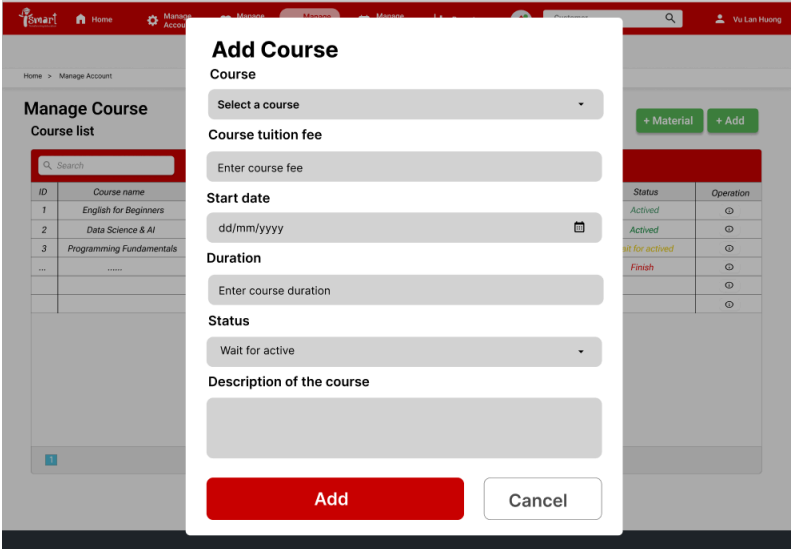
Name	Description	Type	Min	Max	Maxlength	Required
course_name	Name of the course	varchar(100)	-	-	100	yes

	<table><tr><td>description</td><td>Description of the course</td><td>text</td><td>-</td><td>-</td><td>500</td><td>no</td></tr><tr><td>duration</td><td>Course duration</td><td>number</td><td>1</td><td>999</td><td>-</td><td>yes</td></tr><tr><td>fee</td><td>Course tuition fee</td><td>Decimal</td><td>0</td><td>-</td><td>-</td><td>yes</td></tr><tr><td>status</td><td>Course status</td><td>varchar</td><td>-</td><td>-</td><td>20</td><td>yes</td></tr><tr><td>created_at</td><td>Created date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>System</td></tr></table>	description	Description of the course	text	-	-	500	no	duration	Course duration	number	1	999	-	yes	fee	Course tuition fee	Decimal	0	-	-	yes	status	Course status	varchar	-	-	20	yes	created_at	Created date	datetime	-	-	-	System
description	Description of the course	text	-	-	500	no																														
duration	Course duration	number	1	999	-	yes																														
fee	Course tuition fee	Decimal	0	-	-	yes																														
status	Course status	varchar	-	-	20	yes																														
created_at	Created date	datetime	-	-	-	System																														
Acceptance Criteria	AC1 – View Course List Given Admin opens Course Management page When the system loads Then system displays list of courses AC2 – Display Course Information Given course list is displayed Then each course shows course name, duration, fee, and created date AC3 – View Course Details Given Admin is on course list When Admin selects a course Then system displays course details AC4 – Search Course Given Admin enters keyword When performing search Then system displays matching courses AC5 – No Course Available Given no courses exist When page loads																																			

	Then system shows message as defined in Business Rules AC6 – No Search Result Given no courses match search When search is performed Then system shows message as defined in Business Rules AC7 – Navigate to Create Course Given Admin clicks “Add New” Then system navigates to Create Course screen AC8 – System Error Handling Given system fails to load data When page loads Then system shows error as defined in Business Rules
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2. USER STORY: Admin Create Courses

Card	- As an Admin - I want to create a new course - So that the system can store course information for future class creation
Summary	This user story allows Admin to create a new course by entering required information such as course name, duration, and fee, and status.
In-Scope:	- Input and validate new course details (name, duration, fee, operational status).

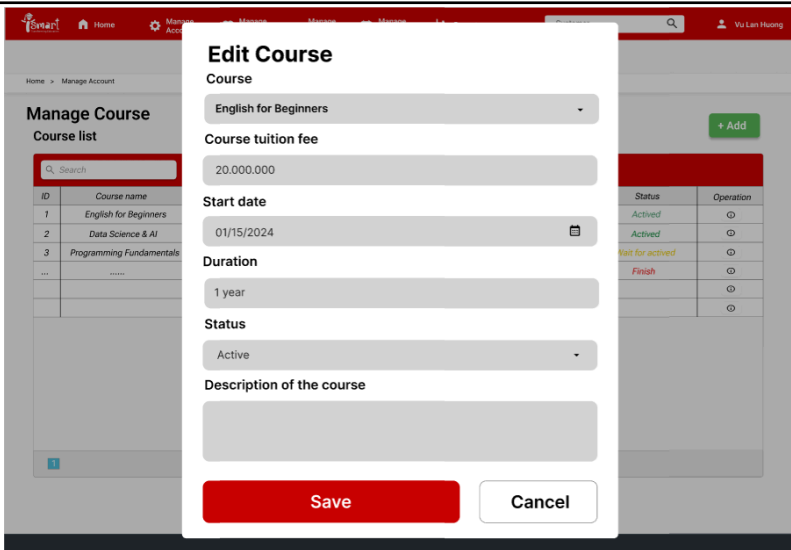
Out of scope	- Assign classes or learning materials directly.
Trigger	Admin clicks the [Add] button on the Course Management screen.
Pre-condition	- Admin must be logged into the system - Admin role must have permission to manage courses - System database must be available
Business rule and error message	- Course name must not be empty Error: Course name is required - Course name must be unique Error: Course name already exists. - Duration must be greater than 0 Error: <i>Duration must be greater than 0.</i> - Fee must be a valid positive number Error: <i>Course fee must be greater than 0.</i> - System must successfully save data Error: <i>Unable to save course data.</i> - Course status must be selected Error: "Course status is required."
Screen design	
Screen description	Create Course screen allows Admin to enter course information.

	Then system shows error as defined in Business Rules AC5 – Invalid Duration Given Admin enters duration ≤ 0 When submitting form Then system shows error as defined in Business Rules AC6 – Invalid Fee Given Admin enters fee ≤ 0 When submitting form Then system shows error as defined in Business Rules AC7 – Status Selection Required Given Admin does not select a status When submitting form Then system shows error as defined in Business Rules. AC8 – Cancel Create Course Given Admin is on Create Course screen When Admin clicks Cancel Then system returns to Course List And no data is saved AC9 – Data Persistence Given course is created successfully When Admin returns to Course List Then new course appears in the list AC10 – System Error Handling Given system fails to save data When Admin submits form Then system shows error as defined in Business Rules
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3. USER STORY: Admin Update Courses

Card	- As an Admin - I want to update course information and its operational status - So that courses data remains accurate and up to date
Summary	This user story allows Admin to edit existing course information such as course name, description, duration, and fee and its operational status.
In-Scope:	- Modify existing course details and toggle its operational status.
Out of scope	- Delete the course
Trigger	Admin clicks the [Edit] icon on a specific course record.
Pre-condition	- Admin is logged in. - Admin must have permission to manage classes - At least one course must exist in the system - System database must be available
Business rule and error message	- Course name must not be empty Error: "Course name is required." - Course name must be unique (excluding current course) Error: "Course name already exists." - Duration must be greater than 0 Error: "Duration must be greater than 0." - Fee must be a positive number Error: "Course fee must be greater than 0." - System must successfully update data Error: "Unable to update course data."

Screen design



Screen description

Update Course screen allows Admin to modify existing course information.

Includes:

- Course Name input field (editable)
- Description textarea
- Duration input field
- Fee input field
- Status dropdown
- **Save button** (update course)
- **Cancel button** (return to Course List)

Name	Description	Type	Min	Max	Maxlength	Required
course_name	Name of the course	varchar(100)	-	-	100	yes
description	Description of the course	text	-	-	500	no
duration	Course duration	number	1	999	-	yes
fee	Course tuition fee	Decimal	1	-	-	yes

	<table><tr><td>status</td><td>Course status</td><td>varchar</td><td>-</td><td>-</td><td>20</td><td>yes</td></tr></table>	status	Course status	varchar	-	-	20	yes
status	Course status	varchar	-	-	20	yes		
Acceptance Criteria	AC1 – Open Update Course Screen Given Admin is on Course Management page When Admin clicks Edit on a course Then system displays Update Course screen And shows existing course data AC2 – Update Course Successfully Given Admin updates valid course information When Admin clicks Save Then system updates course successfully And redirects to Course List AC3 – Course Name Required Given Admin clears course_name When submitting form Then system shows error as defined in Business Rules AC4 – Duplicate Course Name Given another course already has the same name When Admin submits form Then system shows error as defined in Business Rules AC5 – Invalid Duration Given Admin enters duration ≤ 0 When submitting form							

Then system shows error as defined in Business Rules

AC6 – Invalid Fee

Given Admin enters fee ≤ 0

When submitting form

Then system shows error as defined in Business Rules

AC7 – Change Course Status

Given Admin changes the status from Active to Inactive (or vice versa)

When Admin clicks Save

Then system updates the status and reflects it in the Course List.

AC8 – Cancel Update

Given Admin is on Update Course screen

When Admin clicks Cancel

Then system returns to Course List

And no changes are saved

AC9 – Data Persistence After Update

Given course is updated successfully

When Admin returns to Course List

Then updated information is displayed

AC10 – System Error Handling

Given system fails to update data

When Admin submits form

Then system shows error as defined in Business Rules

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4. USER STORY: Admin Delete Courses

Card	- As an Admin - I want to delete a course - So that outdated or unused courses are removed from the system
Summary	This user story allows Admin to delete a course from the system, with validation to ensure no related classes exist.
In-Scope:	- Delete a course with strict constraint validation (blocked if classes exist).
Out of scope	- Recover deleted courses.
Trigger	Admin clicks the [Delete] icon on a specific course record.
Pre-condition	- Admin must be logged into the system - Admin role must have permission to manage courses - System database must be available
Business rule and error message	- Admin must confirm before deletion Message: "Are you sure you want to delete this course?" - Course cannot be deleted if there are classes associated Error: "Course cannot be deleted because classes are associated with it." - System must successfully delete course Error: "Unable to delete course."

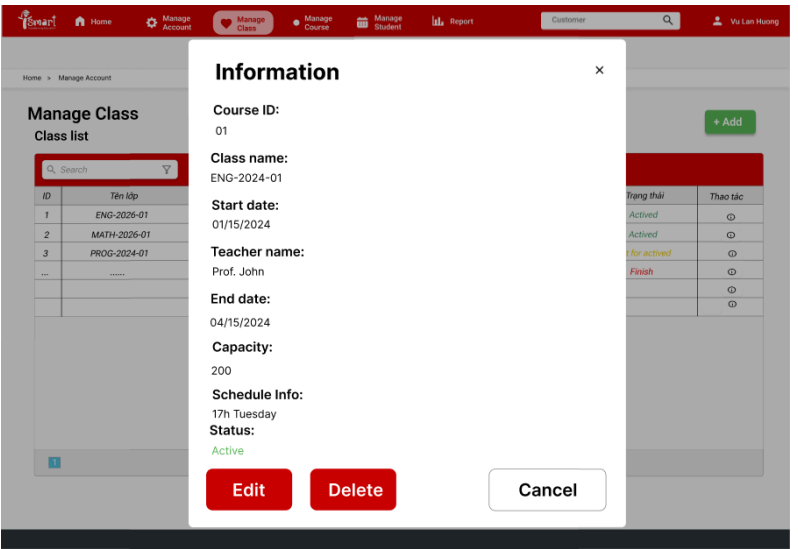
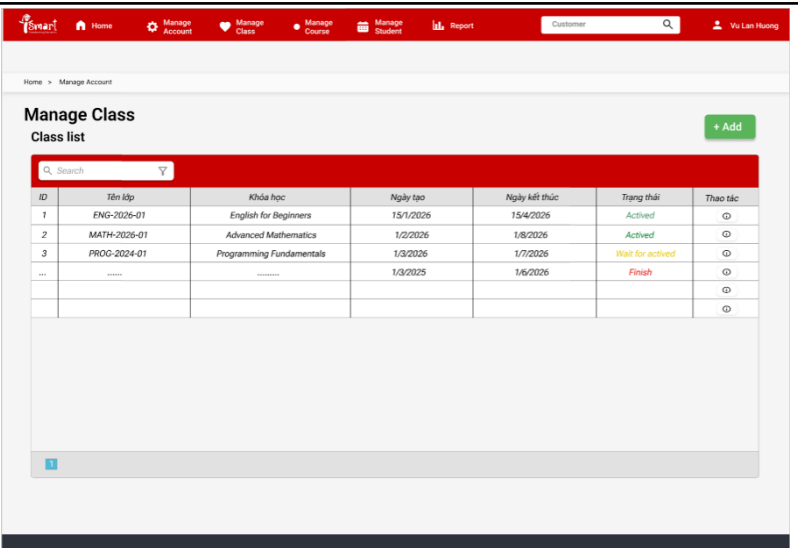
	Given Admin is on Course Management page When Admin clicks Delete on a course Then system displays confirmation dialog AC2 – Confirm Deletion Given confirmation dialog is displayed When Admin clicks Confirm Then system proceeds with deletion AC3 – Cancel Deletion Given confirmation dialog is displayed When Admin clicks Cancel Then system closes dialog And no data is deleted AC4 – Delete Course Successfully Given course has no associated classes When Admin confirms deletion Then system deletes course And removes it from the list AC5 – Delete Course Failed (Constraint) Given course has associated classes When Admin attempts to delete Then system shows error as defined in Business Rules
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	AC6 – Data Persistence After Delete Given course is deleted successfully When course list reloads Then deleted course no longer appears AC7 – System Error Handling Given system fails to delete course When Admin confirms deletion Then system shows error as defined in Business Rules AC8 – UI Update After Delete Given deletion is successful When system updates UI Then course list refreshes automatically
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5. USER STORY: Admin View Classes

Card	- As an Admin
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	- I want to view and search classes including viewing class details - So that I can monitor and access full course information
Summary	This user story allows Admin to view the list of classes, search/filter classes, and access detailed information of each class.
In-Scope:	- View class list, view detailed information, search by name, and filter by course.
Out of scope	- Modify class data.
Trigger	Admin clicks the [Manage Class] menu on the Sidebar.
Pre-condition	- Admin must be logged into the system - Admin role must have permission to manage courses - System database must be available
Business rule and error message	-System must retrieve class data from database Error: "Unable to load class data" If no classes exist in the system Message: "No classes available" If search/filter returns no result Message: "No classes found"
Screen design	



Screen description

Class Management screen allows Admin to view and search class information.

Name	Description	Type	Min	Max	Maxlength	Required
course_id	Course that class belongs to	int	-	-	-	Yes
class_name	Name of class	varchar	1	100	100	Yes
teacher_id	assigned teacher	varchar	1	100	100	no

	AC4 – Search Class Given Admin enters keyword When performing search Then system displays matching classes AC5 – Filter Class by Course Given Admin selects a course in filter When filter is applied Then system displays classes belonging to that course AC6 – No Class Available Given no classes exist When page loads Then system shows message as defined in Business Rules AC7 – No Search/Filter Result Given no classes match search/filter When action is performed Then system shows message as defined in Business Rules AC8 – Navigate to Create Class
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	Given Admin clicks “Add” Then system navigates to Create Class screen AC9 – System Error Handling Given system fails to load class data When page loads Then system shows error as defined in Business Rules
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6. USER STORY: Admin Create Classes

Card	- As an Admin - I want to create a new class - So that courses can have scheduled classes for students
Summary	This user story allows Admin to create a new class by assigning it to a course and providing necessary information such as class name, teacher, start date, capacity, and schedule information.
In-Scope:	- Input class details, schedule info, and assign the class to a specific course.
Out of scope	
Trigger	Admin clicks the [Add New] button on the Class Management screen.
Pre-condition	- Admin must be logged into the system - At least one course exists in the system. - System database must be available

Business rule and error message	Class must belong to an existing course Error: <i>"Selected course does not exist."</i> Class name must not be empty Error: <i>"Class name is required."</i> Class name must be unique within the same course Error: <i>"Class name already exists for this course."</i> Teacher name must not be empty Error: <i>"Teacher name is required."</i> Start date must be today or a future date Error: <i>"Start date must be today or a future date."</i> Capacity must be greater than 0 Error: <i>"Capacity must be greater than 0."</i> Schedule info must not be empty Error: <i>"Schedule information is required."</i> System must successfully save class Error: <i>"Unable to create class."</i>
Screen design	

The image shows a web application interface for managing classes. In the foreground, a modal window titled "Add Class" is open. It contains several input fields: "Course ID" with a dropdown menu labeled "Select a course", "Class name" with a text input field showing "Eg: ENG-2024-01", "Teacher name" with a text input field showing "Prof. John", "Start date" and "End date" with date pickers showing "dd/mm/yyyy", "Schedule Info" with a text input field, "Capacity" with a text input field showing "200", and "Status" with a dropdown menu showing "Wait to active". At the bottom of the modal are two buttons: a red "Create" button and a white "Cancel" button. In the background, a "Manage Class" screen is visible, showing a table with columns "ID", "Tên lớp", and "Trạng thái". The table has three rows of data. There is also a search bar and a "+ Add" button on the right side of the background screen.

Screen description

Create Class screen allows Admin to enter class information.

Includes:

- **Course dropdown** (select course)
- Class Name input field
- Teacher Name input field
- Start Date picker
- Capacity input field
- Schedule Info input field
- **Create button** (submit form)
- **Cancel button** (return to Class List)

Name	Description	Type	Min	Max	Maxlength	Required
course_id	Course that class belongs to	int	-	-	-	Yes
class_name	Name of class	varchar	1	100	100	Yes
teacher_id	assigned teacher	varchar	1	100	100	no

AC4 – Duplicate Class Name in Course

Given class name already exists in selected course
When submitting form
Then system shows error as defined in Business Rules

AC5 – Invalid Course Selection

Given selected course does not exist
When submitting form
Then system shows error as defined in Business Rules

AC6 – Invalid Start Date

Given Admin selects a past date
When submitting form
Then system shows error as defined in Business Rules

AC7 – Invalid Capacity

Given Admin enters capacity ≤ 0
When submitting form
Then system shows error as defined in Business Rules

AC8 – Schedule Info Required

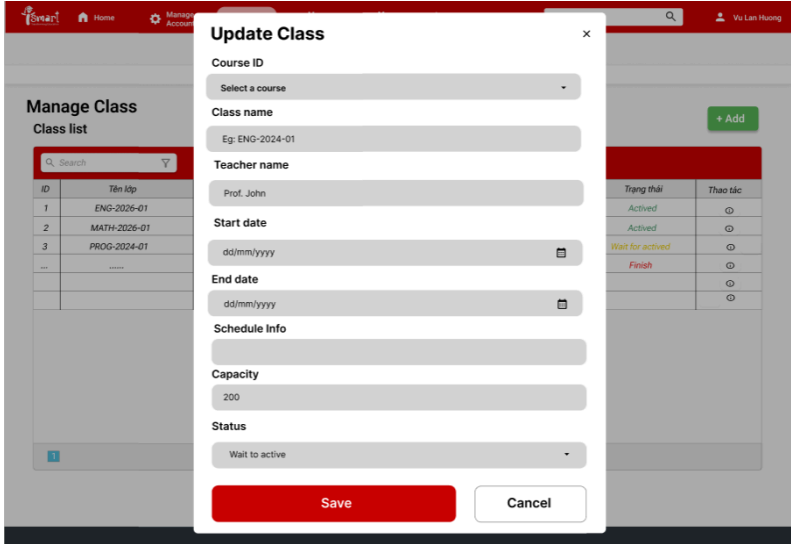
Given Admin leaves schedule_info empty

	When submitting form Then system shows error as defined in Business Rules. AC9 – Cancel Create Class Given Admin is on Create Class screen When Admin clicks Cancel Then system returns to Class List And no data is saved AC10 – System Error Handling Given system fails to save class When submitting form Then system shows error as defined in Business Rules
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7. USER STORY: Admin Update Classes

Card	- As an Admin - I want to update a new class - So that class data remains accurate and classes can be activated for student enrollment.
Summary	This user story allows Admin to edit existing class information such as class name, assigned course, teacher, start date, capacity, schedule info, and change the class status.
In-Scope:	- Modify existing class details, schedule, capacity, and change operational status.

Out of scope	- Delete the class.
Trigger	Admin clicks the [Edit] icon on a specific class record.
Pre-condition	- Admin must be logged into the system - Admin role must have permission to manage courses - System database must be available
Business rule and error message	Class must belong to an existing course Error: <i>"Selected course does not exist."</i> Class name must not be empty Error: <i>"Class name is required."</i> Class name must be unique within the same course Error: <i>"Class name already exists for this course."</i> Teacher name must not be empty Error: <i>"Teacher name is required."</i> Start date must be today or a future date Error: <i>"Start date must be today or a future date."</i> Capacity must be greater than 0 Error: <i>"Capacity must be greater than 0."</i> Schedule info must not be empty (Đã thêm) Error: <i>"Schedule information is required."</i> System must successfully save class Error: <i>"Unable to create class."</i>

Screen design																						
Screen description	Update Class screen allows Admin to modify existing class information. Includes: ● Course dropdown (editable) ● Class Name input field ● Teacher dropdown ● Start Date picker ● Capacity input field ● Save button (update class) ● Cancel button (return to Class List) <table> <tr> <th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr> <tr> <td>course_id</td><td>Course that class belongs to</td><td>int</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr> <tr> <td>class_name</td><td>Name of class</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>Yes</td></tr> </table>	Name	Description	Type	Min	Max	Maxlength	Required	course_id	Course that class belongs to	int	-	-	-	Yes	class_name	Name of class	varchar	1	100	100	Yes
Name	Description	Type	Min	Max	Maxlength	Required																
course_id	Course that class belongs to	int	-	-	-	Yes																
class_name	Name of class	varchar	1	100	100	Yes																

Given Admin clears class_name
When submitting form
Then system shows error as defined in Business Rules

AC4 – Duplicate Class Name in Course

Given another class in the same course has the same name
When submitting form
Then system shows error as defined in Business Rules

AC5 – Invalid Course Selection

Given selected course does not exist
When submitting form
Then system shows error as defined in Business Rules

AC6 – Invalid Start Date

Given Admin selects a past date
When submitting form
Then system shows error as defined in Business Rules

AC7 – Invalid Capacity

Given Admin enters capacity ≤ 0
When submitting form

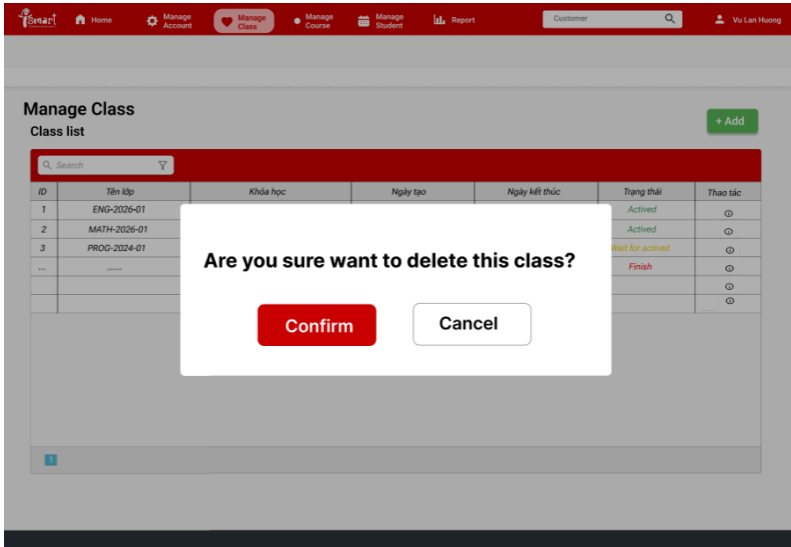
	Then system shows error as defined in Business Rules AC8 – Schedule Info Required Given Admin clears schedule_info When submitting form Then system shows error as defined in Business Rules. AC9 – Activate Class / Update Status Given class status is currently "Wait to active" When Admin changes Status to "Active" and clicks Save Then system updates the status successfully and makes the class available for enrollment operations. AC10 – Cancel Update Given Admin is on Update Class screen When Admin clicks Cancel Then system returns to Class List And no changes are saved – System Error Handling Given system fails to update class When submitting form
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	Then system shows error as defined in Business Rules
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8. USER STORY: Admin Delete Classes

Card	- As an Admin - I want to delete a class - So that unused or invalid classes are removed from the system
Summary	This user story allows Admin to delete a class, with validation to ensure no students are enrolled in the class.
In-Scope:	- Delete a class with strict constraint validation (blocked if students are enrolled).
Out of scope	- Recover deleted classes.
Trigger	Admin clicks Delete button on a class in Class Management screen
Pre-condition	- Admin must be logged into the system - Admin role must have permission to manage courses - System database must be available
Business rule and error message	Admin must confirm before deletion Message: "Are you sure you want to delete this class?" Class cannot be deleted if students are enrolled Error: "Class cannot be deleted because students are enrolled." System must successfully delete class Error: "Unable to delete class."

Screen design



The screenshot shows a web application interface for managing classes. At the top is a navigation bar with links for Home, Manage Account, Manage Class (active), Manage Course, Manage Student, and Report. Below the navigation bar is the 'Manage Class' section, which includes a 'Class list' table and a '+ Add' button. The table has columns for ID, Tên lớp (Class Name), Khóa học (Course), Ngày tạo (Created Date), Ngày kết thúc (End Date), Trạng thái (Status), and Thao tác (Action). A confirmation dialog is overlaid on the table, asking 'Are you sure want to delete this class?' with 'Confirm' and 'Cancel' buttons.

Screen description

Class Management screen allows Admin to delete a class.

Includes:

- Class table with list of classes
- **Delete button** for each class
- **Confirmation dialog:**
 - Confirm button
 - Cancel button

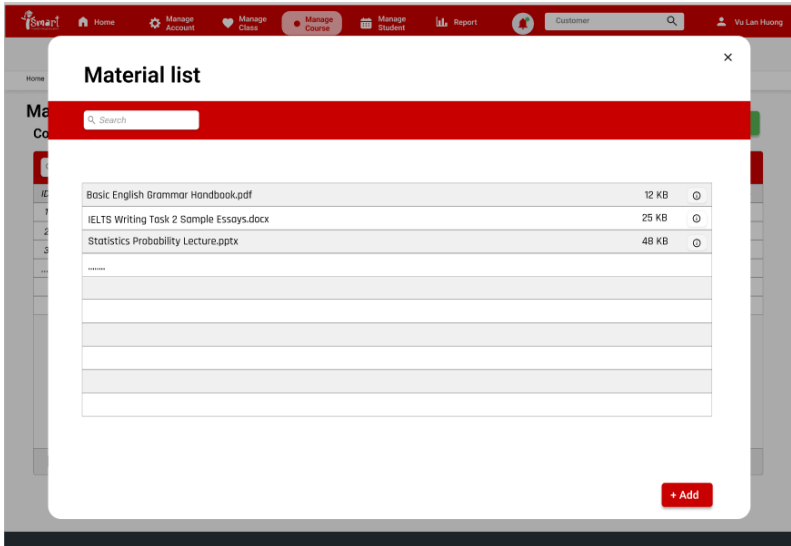
Name	Description	Type	Min	Max	Maxlength	Required
course_id	Course that class belongs to	int	-	-	-	Yes
class_name	Name of class	varchar	1	100	100	Yes

	Then system closes dialog And no data is deleted AC4 – Delete Class Successfully Given class has no students enrolled When Admin confirms deletion Then system deletes class And removes it from the list AC5 – Delete Class Failed (Constraint) Given class has students enrolled When Admin attempts to delete Then system shows error as defined in Business Rules AC6 – Data Persistence After Delete Given class is deleted successfully When class list reloads Then deleted class no longer appears AC7 – System Error Handling Given system fails to delete class When Admin confirms deletion Then system shows error as defined in Business Rules
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	AC8 – UI Update After Delete Given deletion is successful When system updates UI Then class list refreshes automatically
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9. USER STORY: Admin View Material

Card	- As an Admin - I want to view the list of materials for a specific course - So that I can manage and review learning resources associated with that course.
Summary	This user story allows Admin to view all materials within a specific course, including searching by material title and viewing material details.
In-Scope:	- View and search materials strictly associated with a specific course.
Out of scope	
Trigger	Admin clicks the specific course and [Add] button Materail
Pre-condition	- Admin must be logged into the system - Materials may or may not exist in the selected course. - System database must be available
Business rule and error message	- System must retrieve material data from database for the currently selected course only. Error: "Unable to load material data."

	- If no materials exist for this course Message: "No materials found for this course." - Search keyword must match material title Error: System returns matching results within the current course. - If search returns no result Message: "No materials match your search."																																			
Screen design																																				
Screen description	Material Management tab allows Admin to view and search materials within a specific course. <table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>title</td><td>Name of the material</td><td>varchar</td><td>1</td><td>255</td><td></td><td>yes</td></tr><tr><td>file_format</td><td>Format (PDF, PPTX, etc.)</td><td>varchar</td><td>1</td><td>10</td><td>10</td><td>yes</td></tr><tr><td>file_size</td><td>Size of the file</td><td>varchar</td><td>1</td><td>20</td><td>-</td><td>yes</td></tr><tr><td>file_url</td><td>Link to access file</td><td>varchar</td><td>-</td><td>-</td><td></td><td>yes</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	title	Name of the material	varchar	1	255		yes	file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	yes	file_size	Size of the file	varchar	1	20	-	yes	file_url	Link to access file	varchar	-	-		yes
Name	Description	Type	Min	Max	Maxlength	Required																														
title	Name of the material	varchar	1	255		yes																														
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file_url	Link to access file	varchar	-	-		yes																														

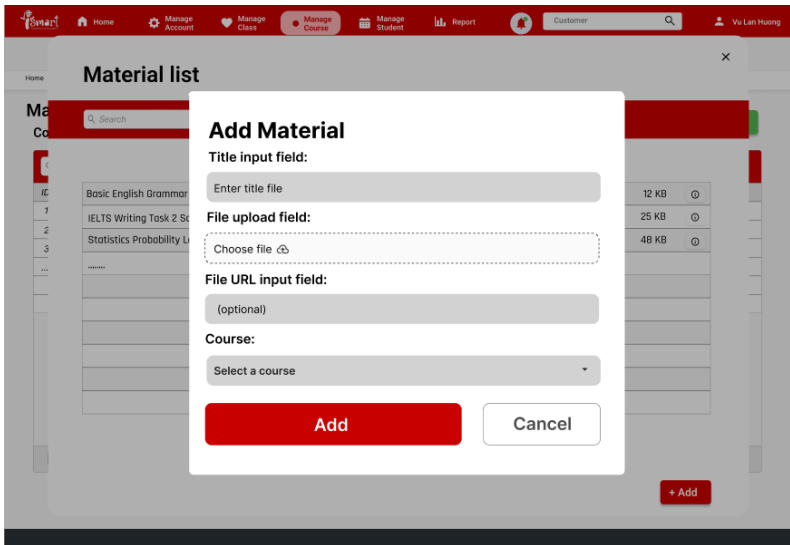
	<table><tr><td>keyword</td><td>Search title</td><td>varchar</td><td>0</td><td>100</td><td>100</td><td>no</td></tr><tr><td>created_at</td><td>Creation date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>yes</td></tr></table>	keyword	Search title	varchar	0	100	100	no	created_at	Creation date	datetime	-	-	-	yes
keyword	Search title	varchar	0	100	100	no									
created_at	Creation date	datetime	-	-	-	yes									
Acceptance Criteria	AC1 – View Contextual Material List Given Admin opens the [All Materials] tab in Course Details When the system loads Then the system displays the list of materials for that specific course. AC2 – Display Material Information Given the material list is displayed Then each material row includes title, file format, file size, and created date. AC3 – View Material Detail Given Admin selects a material When Admin clicks View or selects a row Then the system displays detailed information of the material. AC4 – Search Material Within Course Given Admin enters a keyword When Admin performs search Then the system displays materials matching the keyword within the current course.														

	AC5 – No Material Found (Search) Given no materials match the search keyword When the system returns results Then the system displays message as defined in Business Rules. AC6 – Empty Material State Given there are no materials in the selected course When Admin opens the tab Then the system displays message as defined in Business Rules. AC7 – Data Load Failure Given system fails to retrieve material data When loading the tab Then the system displays error as defined in Business Rules.
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10. USER STORY: Admin Create Material

Card	- As an Admin - I want to create a new material for a specific course - So that learning resources can be added to the course and used for teaching or consultation.
Summary	This user story allows Admin to add new learning materials (either by uploading a file or providing a

	URL) while working within the Material tab of a specific course.
In-Scope:	- Upload file or input URL, auto-extract metadata (format/size), and link to a course.
Out of scope	
Trigger	Admin clicks the specific course and [Add] button Materail
Pre-condition	- Admin must be logged into the system - A specific course is currently selected. - System database must be available
Business rule and error message	- Material title must not be empty. Error: "Material title is required." - Material title length must be between 1 and 255 characters. Message: "Material title must be between 1 and 255 characters." - At least one source must be provided (File OR URL). Error: "Please upload a file or provide a URL." - If a file is uploaded, it must be in supported formats (PDF, DOC, DOCX, PPT, PPTX). Message: "Unsupported file format." - System must successfully save material data. Message: "Unable to create material."

Screen design																																																		
Screen description	Create Material screen (often a Modal/Popup within the Course Detail Hub). <table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>title</td><td>Name of the material</td><td>varchar</td><td>1</td><td>255</td><td></td><td>yes</td></tr><tr><td>file_format</td><td>Format (PDF, PPTX, etc.)</td><td>varchar</td><td>1</td><td>10</td><td>10</td><td>system</td></tr><tr><td>file_size</td><td>Size of the file</td><td>varchar</td><td>1</td><td>20</td><td>-</td><td>system</td></tr><tr><td>file_url</td><td>Link to access file</td><td>varchar</td><td>-</td><td>-</td><td></td><td>system</td></tr><tr><td>created_at</td><td>Creation date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>system</td></tr><tr><td>course_id</td><td>Linked course</td><td>int</td><td>-</td><td>-</td><td>-</td><td>system</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	title	Name of the material	varchar	1	255		yes	file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	system	file_size	Size of the file	varchar	1	20	-	system	file_url	Link to access file	varchar	-	-		system	created_at	Creation date	datetime	-	-	-	system	course_id	Linked course	int	-	-	-	system
Name	Description	Type	Min	Max	Maxlength	Required																																												
title	Name of the material	varchar	1	255		yes																																												
file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	system																																												
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file_url	Link to access file	varchar	-	-		system																																												
created_at	Creation date	datetime	-	-	-	system																																												
course_id	Linked course	int	-	-	-	system																																												
Acceptance Criteria	AC1 – Open Create Material Screen Given Admin is in the Material tab of a Course When Admin clicks "Add New Material"																																																	

	Then system displays the Create Material screen/modal. AC2 – Create Material Successfully Given Admin provides valid title and a file (or URL) When Admin clicks Add Then system saves the material, extracts file_format and file_size, and associates it with the current course. AC3 – Title Validation Given Admin leaves the title empty or exceeds 255 characters When submitting Then system shows error as defined in Business Rules. AC4 – Conditional Source Validation Given both File upload and URL fields are empty When submitting Then system shows: "Please upload a file or provide a URL." AC5 – File Format Validation Given Admin uploads a file with an unsupported extension (e.g., .exe, .png) When submitting
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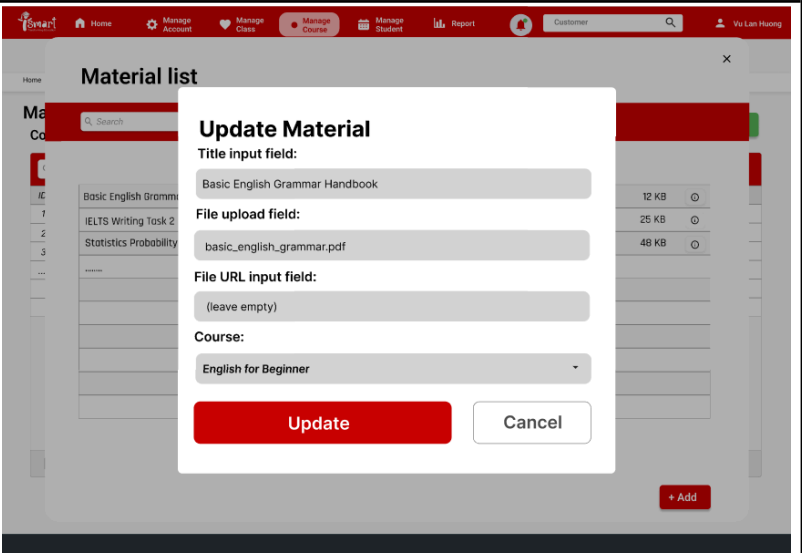
	Then system shows error: "Unsupported file format." AC6 – Automatic Metadata Extraction Given a successful file upload Then system must automatically identify the format (e.g., "PDF") and size (e.g., "4.5 MB") to store in the database. AC7 – Cancel Action Given Admin is on the Create Material screen When Admin clicks Cancel Then system closes the screen and no data is saved.
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11. USER STORY: Admin Update Material

Card	- As an Admin - I want to update material information for a specific course - So that learning resources remain accurate, up-to-date, and correctly formatted for all users.
Summary	This user story allows Admin to modify existing material details, replace files, or update URLs while staying within the context of the specific course's material tab.
In-Scope:	- Replace file/URL, update title, and auto-re-extract metadata.
Out of scope	- Delete the material.

Trigger	Admin clicks the [Edit] icon on a specific material record.
Pre-condition	- Admin must be logged into the system - The material exists and is associated with the currently viewed course. - System database must be available
Business rule and error message	- Material title must not be empty. Error: "Material title is required." - Material title length must be between 1 and 255 characters. Message: "Material title must be between 1 and 255 characters." - At least one source must be provided (File OR URL). Error: "Please upload a file or provide a URL." - If a file is uploaded, it must be in supported formats (PDF, DOC, DOCX, PPT, PPTX). Message: "Unsupported file format." - System must successfully save material data. Message: "Unable to update material."

Screen design



Screen description

Update Material screen (Modal/Popup).

Name	Description	Type	Min	Max	Maxlength	Required
title	Name of the material	varchar	1	255		yes
file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	yes
file_size	Size of the file	varchar	1	20	-	yes
file_url	Link to access file	varchar	-	-		yes
keyword	Search title	varchar	0	100	100	no
created_at	Creation date	datetime	-	-	-	yes

Acceptance Criteria

AC1 – Open Update Screen with Data

Given Admin clicks Edit on a material

Then system displays Update Material modal with all current information pre-filled.

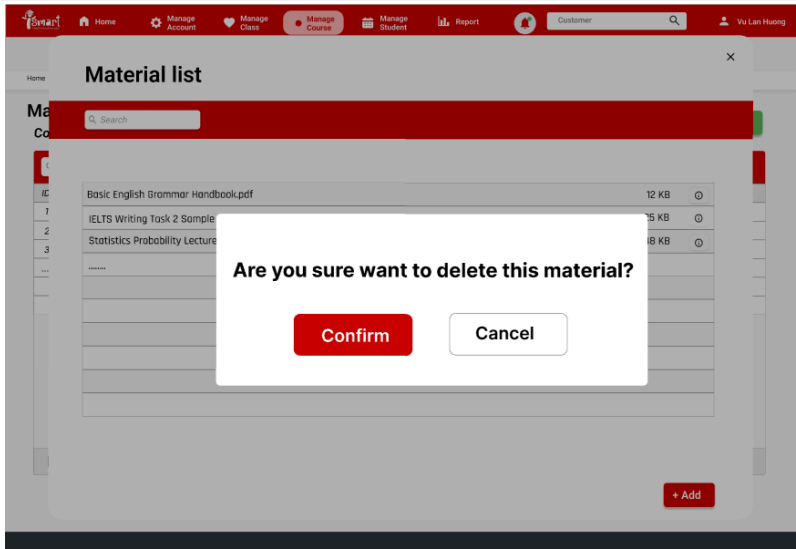
AC2 – Update Title Successfully

	Given Admin modifies the title and clicks Save Then system updates the title and keeps other file data unchanged. AC3 – Replace File Successfully Given Admin uploads a new file When Admin clicks Save Then system replaces the old file, re-extracts the new file_format and file_size, and updates the database. AC4 – Switch from File to URL Given Admin removes the existing file and provides a new URL When Admin clicks Save Then system updates the source to the URL and sets file_format to "Link" (or similar). AC5 – Title Validation Given Admin clears the title field When submitting Then system shows error: "Material title is required." AC6 – File/Source Persistence Given Admin attempts to remove both the file and the URL When submitting
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	Then system shows error as defined in Business Rules. AC7 – Cancel Changes Given Admin makes changes but clicks Cancel Then the modal closes and the material remains in its original state. AC8 – System Error Handling Given system fails to update the database When submitting Then system shows error as defined in Business Rules.
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12. Admin Delete Material

Card	- As an Admin - I want to delete a material - So that outdated or incorrect learning resources can be removed from the system
Summary	This user story allows Admin to delete an existing material from the system while ensuring data consistency
In-Scope:	- Permanently remove a material from a course.

Out of scope	- Recover deleted materials.
Trigger	Admin clicks the [Delete] icon on a specific material record.
Pre-condition	- Admin is logged into the system - Material exists in the system - System database is available
Business rule and error message	- Material must exist in the system → Error: "Material does not exist." - System must confirm before deletion → Message: "Are you sure you want to delete this material?" - Material deletion must be permanent (hard delete) - System must successfully delete material → Error: "Unable to delete material."
Screen design	 The screenshot shows a web application interface for 'Material Management'. At the top, there is a navigation bar with links like Home, Manage Account, Manage Class, Manage Course, Manage Student, Report, and a user profile for 'Vu Lan Huong'. Below the navigation bar is a 'Material list' section with a search bar. A table lists materials with columns for material name, size, and an action icon. The first row shows 'Basic English Grammar Handbook.pdf' with a size of '12 KB'. A confirmation dialog box is overlaid on the table, asking 'Are you sure want to delete this material?' with 'Confirm' and 'Cancel' buttons. At the bottom right of the table, there is a '+ Add' button.
Screen description	Material Management screen includes:

	<table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>title</td><td>Name of the material</td><td>varchar</td><td>1</td><td>255</td><td></td><td>yes</td></tr><tr><td>file_format</td><td>Format (PDF, PPTX, etc.)</td><td>varchar</td><td>1</td><td>10</td><td>10</td><td>yes</td></tr><tr><td>file_size</td><td>Size of the file</td><td>varchar</td><td>1</td><td>20</td><td>-</td><td>yes</td></tr><tr><td>file_url</td><td>Link to access file</td><td>varchar</td><td>-</td><td>-</td><td></td><td>yes</td></tr><tr><td>keyword</td><td>Search title</td><td>varchar</td><td>0</td><td>100</td><td>100</td><td>no</td></tr><tr><td>created_at</td><td>Creation date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>yes</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	title	Name of the material	varchar	1	255		yes	file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	yes	file_size	Size of the file	varchar	1	20	-	yes	file_url	Link to access file	varchar	-	-		yes	keyword	Search title	varchar	0	100	100	no	created_at	Creation date	datetime	-	-	-	yes
Name	Description	Type	Min	Max	Maxlength	Required																																												
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keyword	Search title	varchar	0	100	100	no																																												
created_at	Creation date	datetime	-	-	-	yes																																												
Acceptance Criteria	AC1 – Display Delete Option Given material list is displayed Then each material has a Delete button AC2 – Trigger Delete Action Given Admin clicks Delete on a material When action is triggered Then system displays confirmation dialog AC3 – Confirm Delete Successfully Given Admin confirms deletion When system processes request Then material is removed from the system And no longer appears in the list AC4 – Cancel Delete Given confirmation dialog is displayed When Admin clicks Cancel Then system closes dialog And no data is deleted																																																	

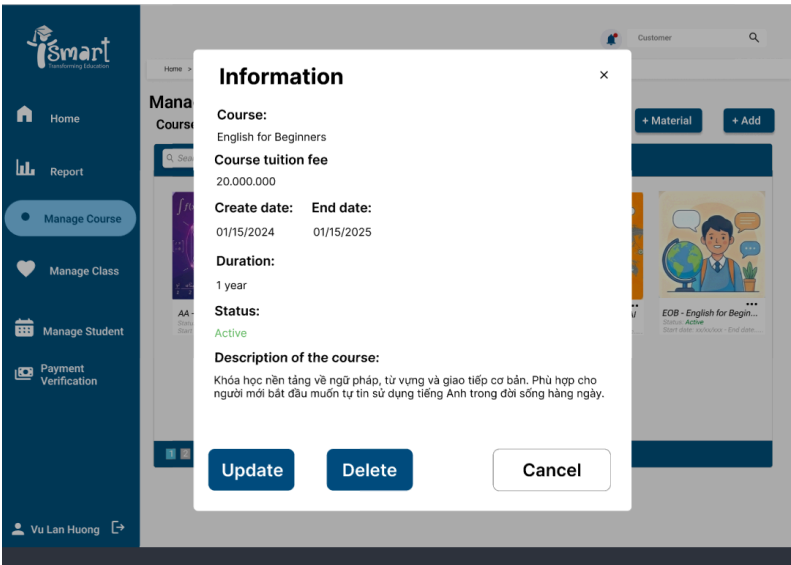
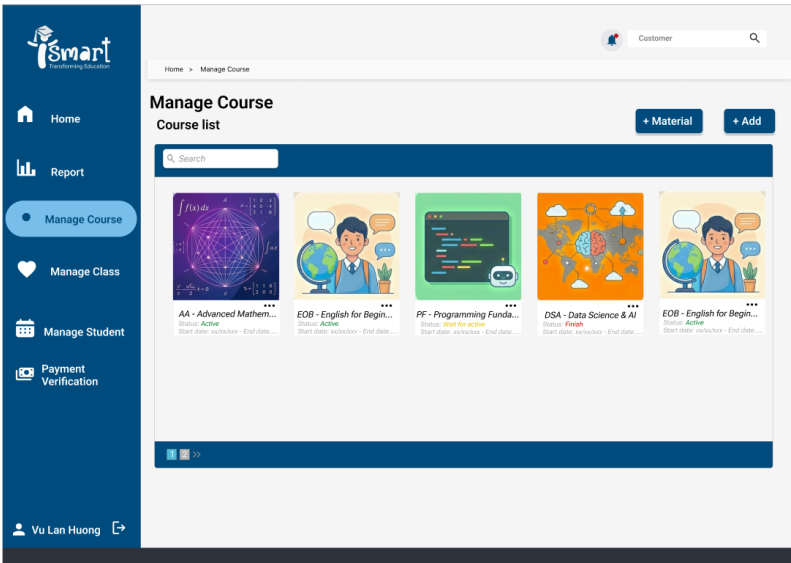
	AC5 – Material Not Found Given material does not exist When Admin attempts to delete Then system shows error as defined in Business Rules AC6 – System Error Handling Given system fails to delete material When Admin confirms deletion Then system shows error as defined in Business Rules AC7 – Refresh Material List After Delete Given material is deleted successfully When system updates UI Then material list is refreshed And deleted item is not displayed AC8 – Multiple Delete Attempts Given material has already been deleted When Admin attempts to delete again Then system shows error as defined in Business Rules AC9 – Delete Without Affecting Other Data Given material is deleted When system processes deletion Then no other materials or course data are affected
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Staff

1. USER STORY: Staff View Course

Card	- As a Staff - I want to view and search courses - So that I can access course information for operations or consultation
Summary	This user story allows users to view the list of courses, search courses, and view detailed course information
In-Scope:	- View the course list, search by keyword, and view specific course details.
Out of scope	- Modify course data.
Trigger	Staff clicks the [Manage Course] menu on the Sidebar.
Pre-condition	- User is logged in - Courses exist in the system - System database is available
Business rule and error message	- System must retrieve course data from database → Error: "Unable to load course data." - Search keyword must be applied to course name → Behavior: return matching results - If no course matches search result → Message: "No courses found." - Session must be valid → Error: "Session expired. Please log in again."

Screen design



Screen description

Course List screen allows User to view and search courses.

Name	Description	Type	Min	Max	Maxlength	Required
course_name	Name of the course	varchar(100)	-	-	100	yes
description	Description of the course	text	-	-	500	no
duration	Course duration	number	1	999	-	yes

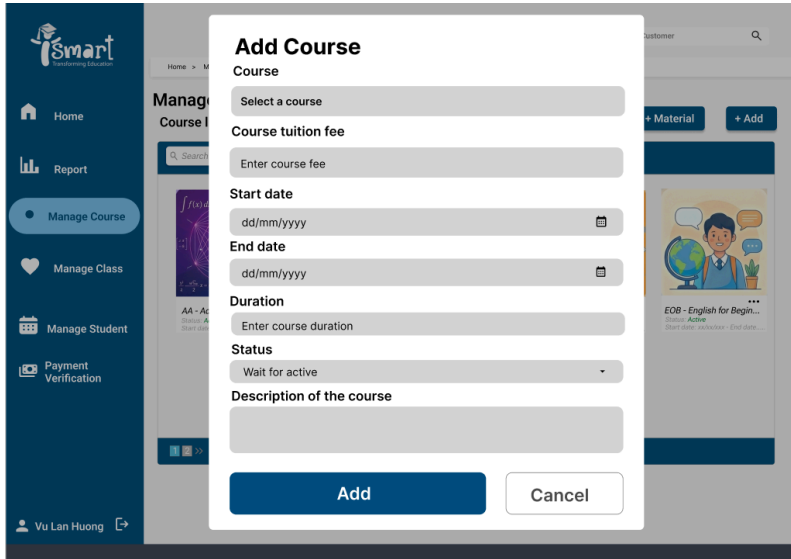
	<table><tr><td>fee</td><td>Course tuition fee</td><td>Decimal</td><td>0</td><td>-</td><td>-</td><td>yes</td></tr><tr><td>status</td><td>Course status</td><td>varchar</td><td>-</td><td>-</td><td>20</td><td>yes</td></tr><tr><td>created_at</td><td>Created date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>System</td></tr></table>	fee	Course tuition fee	Decimal	0	-	-	yes	status	Course status	varchar	-	-	20	yes	created_at	Created date	datetime	-	-	-	System
fee	Course tuition fee	Decimal	0	-	-	yes																
status	Course status	varchar	-	-	20	yes																
created_at	Created date	datetime	-	-	-	System																
Acceptance Criteria	AC1 – View course list Given Staff opens Course List page When system loads Then system displays list of courses AC2 – Display Course Information Given course list is displayed When viewing the list Then each course shows name, status, start date and end date AC3 – View Course Detail Given course list is displayed When Staff clicks a course Then system displays course detail AC4 – Search Course Given Staff enters keyword When Staff clicks Search Then system displays matching courses AC5 – Search Result Accuracy Given keyword matches course name When search is performed Then system returns correct results AC6 – No Course Found																					

	Given no course matches keyword When search is performed Then system shows message as defined in Business Rules AC7 – Empty Course List Given no courses exist in system When Staff opens Course List Then system shows message as defined in Business Rules AC8 – Session Expired Given session is expired When Staff accesses Course List Then system redirects to login page And shows error as defined in Business Rules AC9 – System Error Handling Given system fails to retrieve data When loading course list Then system shows error as defined in Business Rules
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2. USER STORY: Staff Create Course

Card	- As an Staff - I want to create a new course - So that courses can be added for class organization and material assignment
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Summary	This user story allows Staff to create a new course by entering required course information into the system
In-Scope:	- Input and validate new course details (name, duration, fee, operational status).
Out of scope	- Assign classes or learning materials directly.
Trigger	Staff clicks the [Add] button on the Course Management screen.
Pre-condition	- Staff is logged in - Staff has permission to create course - System database is available
Business rule and error message	- Course name must not be empty → Error: "Course name is required." - Course name must be unique in the system → Error: "Course name already exists." - Duration must be greater than 0 → Error: "Duration must be greater than 0." - Fee must be a valid positive number → Error: "Course fee must be greater than 0."

	- System must successfully create course → Error: "Unable to create course."																																																	
Screen design																																																		
Screen description	Create Course screen allows Staff to add new course information. <table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>course_name</td><td>Name of the course</td><td>varchar (100)</td><td>-</td><td>-</td><td>100</td><td>yes</td></tr><tr><td>description</td><td>Description of the course</td><td>text</td><td>-</td><td>-</td><td>500</td><td>no</td></tr><tr><td>duration</td><td>Course duration</td><td>number</td><td>1</td><td>999</td><td>-</td><td>yes</td></tr><tr><td>fee</td><td>Course tuition fee</td><td>Decimal</td><td>0</td><td>-</td><td>-</td><td>yes</td></tr><tr><td>status</td><td>Course status</td><td>varchar</td><td>-</td><td>-</td><td>20</td><td>yes</td></tr><tr><td>created_at</td><td>Created date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>System</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	course_name	Name of the course	varchar (100)	-	-	100	yes	description	Description of the course	text	-	-	500	no	duration	Course duration	number	1	999	-	yes	fee	Course tuition fee	Decimal	0	-	-	yes	status	Course status	varchar	-	-	20	yes	created_at	Created date	datetime	-	-	-	System
Name	Description	Type	Min	Max	Maxlength	Required																																												
course_name	Name of the course	varchar (100)	-	-	100	yes																																												
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created_at	Created date	datetime	-	-	-	System																																												
Acceptance Criteria	AC1 – Open Create Course Screen																																																	

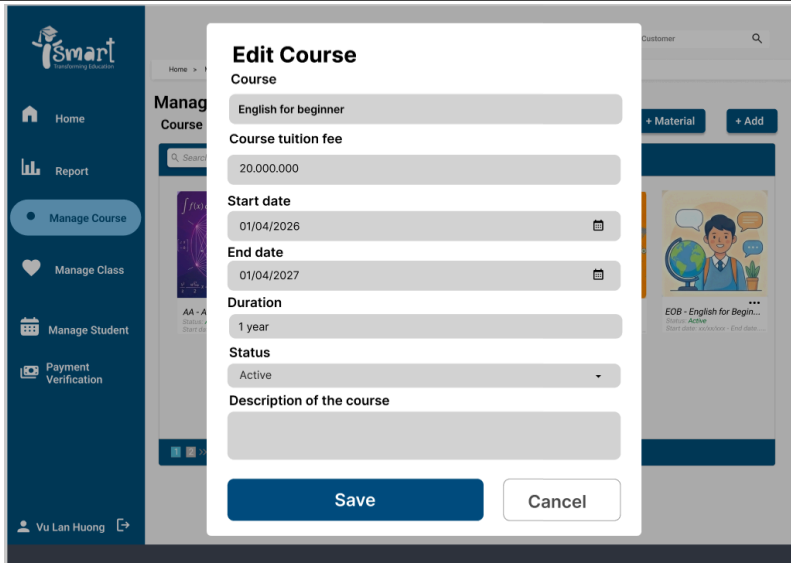
	Given Staff is on Course Management page When Staff clicks Add New Course Then system displays Create Course screen AC2 – Create Course Successfully Given Staff enters valid course information When Staff clicks Add Then system saves course successfully And redirects to Course List AC3 – Course Name Required Given Staff leaves course_name empty When submitting form Then system shows error as defined in Business Rules AC4 – Duplicate Course Name Given a course with the same name already exists When Staff submits form Then system shows error as defined in Business Rules AC5 – Invalid Duration Given Staff enters duration ≤ 0 When submitting form Then system shows error as defined in Business Rules AC6 – Invalid Fee Given Staff enters invalid or negative fee When submitting form
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	Then system shows error as defined in Business Rules AC7 – Status Selection Required Given Staff does not select a status When submitting form Then system shows error as defined in Business Rules. AC8 – Cancel Create Given Staff is on Create Course screen When Staff clicks Cancel Then system returns to Course List And no data is saved AC9 – System Error Handling Given system fails to save course When Staff submits form Then system shows error as defined in Business Rules
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3. USER STORY: Staff Update Course

Card	- As an Staff - I want to update course information - So that course data remains accurate for class organization and material usage
Summary	This user story allows Staff to edit existing course information such as course name, description, duration, and fee

In-Scope:	- Modify existing course details and toggle its operational status.
Out of scope	
Trigger	Admin clicks the [Edit] icon on a specific course record.
Pre-condition	- Staff is logged in - Course exists in the system - Staff has permission to update course - System database is available
Business rule and error message	- Course name must not be empty → Error: "Course name is required." - Course name must be unique (excluding current course) → Error: "Course name already exists." - Duration must be greater than 0 → Error: "Duration must be greater than 0." - Fee must be a valid positive number → Error: "Course fee must be greater than 0." - System must successfully update course → Error: "Unable to update course."
Screen design	

																																																		
Screen description	Update Course screen allows Staff to modify existing course information. <table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>course_name</td><td>Name of the course</td><td>varchar(100)</td><td>-</td><td>-</td><td>100</td><td>yes</td></tr><tr><td>description</td><td>Description of the course</td><td>text</td><td>-</td><td>-</td><td>500</td><td>no</td></tr><tr><td>duration</td><td>Course duration</td><td>number</td><td>1</td><td>999</td><td>-</td><td>yes</td></tr><tr><td>fee</td><td>Course tuition fee</td><td>Decimal</td><td>0</td><td>-</td><td>-</td><td>yes</td></tr><tr><td>status</td><td>Course status</td><td>varchar</td><td>-</td><td>-</td><td>20</td><td>yes</td></tr><tr><td>created_at</td><td>Created date</td><td>date time</td><td>-</td><td>-</td><td>-</td><td>System</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	course_name	Name of the course	varchar(100)	-	-	100	yes	description	Description of the course	text	-	-	500	no	duration	Course duration	number	1	999	-	yes	fee	Course tuition fee	Decimal	0	-	-	yes	status	Course status	varchar	-	-	20	yes	created_at	Created date	date time	-	-	-	System
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course_name	Name of the course	varchar(100)	-	-	100	yes																																												
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created_at	Created date	date time	-	-	-	System																																												
Acceptance Criteria	AC1 – Open Update Course Screen Given Staff is on Course Management page When Staff clicks Edit on a course Then system displays Update Course screen And shows existing course data AC2 – Update Course Successfully																																																	

	Given Staff updates valid course information When Staff clicks Save Then system updates course successfully And redirects to Course List AC3 – Course Name Required Given Staff clears course_name When submitting form Then system shows error as defined in Business Rules AC4 – Duplicate Course Name Given another course with the same name exists When submitting form Then system shows error as defined in Business Rules AC5 – Invalid Duration Given Staff enters duration ≤ 0 When submitting form Then system shows error as defined in Business Rules AC6 – Invalid Fee Given Staff enters invalid or negative fee When submitting form Then system shows error as defined in Business Rules AC7 – Change Course Status Given Staff changes the status from Active to Inactive (or vice versa)
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	When Staff clicks Save Then system updates the status and reflects it in the Course List. AC8 – Cancel Update Given Staff is on Update Course screen When Staff clicks Cancel Then system returns to Course List And no changes are saved AC9 – Data Persistence After Update Given course is updated successfully When returning to Course List Then updated data is displayed correctly AC10 – System Error Handling Given system fails to update course When Staff submits form Then system shows error as defined in Business Rules
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4. USER STORY: Staff Delete Course

Card	- As a Staff - I want to delete a course - So that unused or invalid courses are removed from the system
Summary	This user story allows Staff to delete a course with validation to ensure no classes are associated with it

In-Scope:	- Delete a course with strict constraint validation (blocked if classes exist).
Out of scope	- Recover deleted courses.
Trigger	Staff clicks the [Delete] icon on a specific course record.
Pre-condition	- Staff is logged in - Course exists in the system - Staff has permission to delete course - System database is available
Business rule and error message	- Staff must confirm before deletion → Message: "Are you sure you want to delete this course?" - Course cannot be deleted if there are classes associated with it → Error: "Course cannot be deleted because classes are associated with it." - System must successfully delete course → Error: "Unable to delete course."
Screen design	

The screenshot shows the 'iSmart' web application interface. On the left is a dark blue sidebar with navigation icons and labels: Home, Report, Manage Course (highlighted), Manage Class, Manage Student, and Payment Verification. At the bottom of the sidebar, the user 'Vu Lan Huong' is logged in. The main content area is titled 'Manage Course' and 'Course list'. It features a search bar and two buttons: '+ Material' and '+ Add'. A modal dialog box is centered on the screen, asking 'Are you sure want to delete this course?' with 'Confirm' and 'Cancel' buttons. In the background, a table of courses is partially visible, including one titled 'EOB - English for Begin...'. The user's name 'Vu Lan Huong' is also visible in the top right corner of the application header.

Screen description

Course Management screen allows Staff to delete courses.

Includes:

- Course table
- **Delete button** for each course
- **Confirmation dialog:**
 - Confirm button
 - Cancel button

Name	Description	Type	Min	Max	Maxlength	Required
course_name	Name of the course	varchar(100)	-	-	100	yes
description	Description of the course	text	-	-	500	no
duration	Course duration	number	1	999	-	yes
fee	Course tuition fee	Decimal	0	-	-	yes
status	Course status	varchar	-	-	20	yes
created_at	Created date	datetime	-	-	-	System

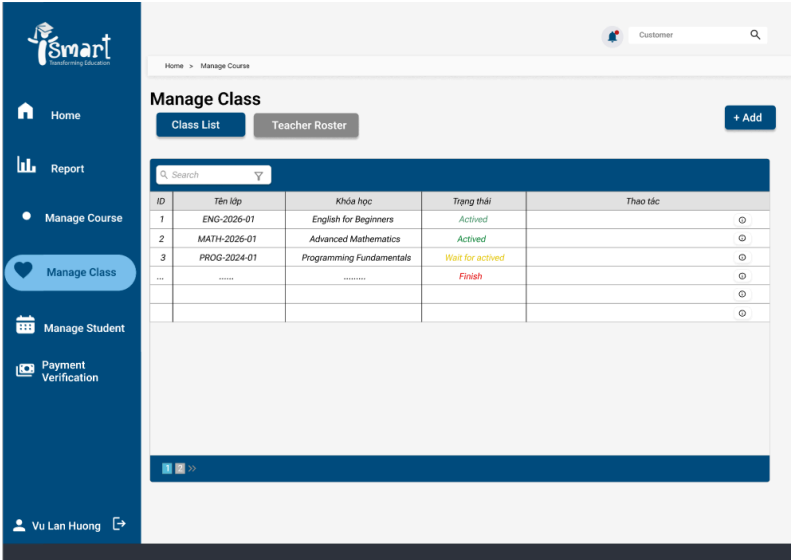
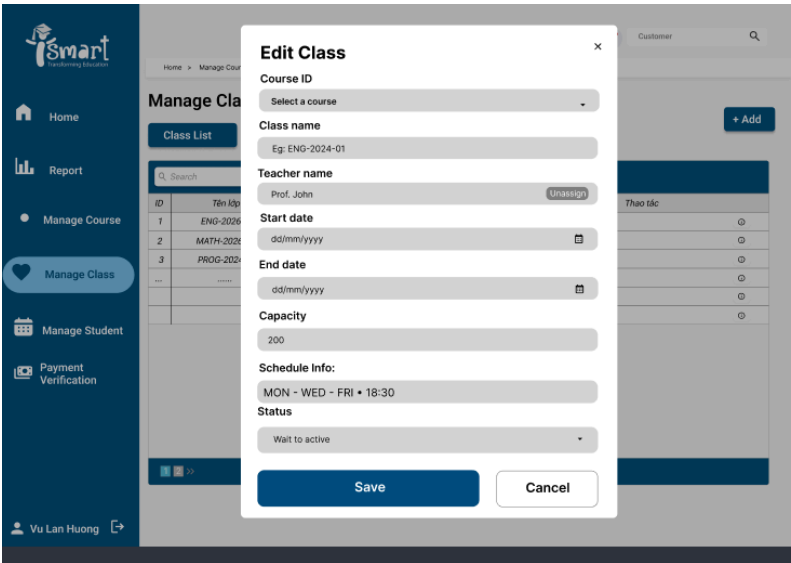
Acceptance Criteria	AC1 – Trigger Delete Action Given Staff is on Course Management page When Staff clicks Delete on a course Then system displays confirmation dialog AC2 – Confirm Deletion Given confirmation dialog is displayed When Staff clicks Confirm Then system proceeds with deletion AC3 – Cancel Deletion Given confirmation dialog is displayed When Staff clicks Cancel Then system closes dialog And no data is deleted AC4 – Delete Course Successfully Given course has no associated classes When Staff confirms deletion Then system deletes course And removes it from the list AC5 – Delete Course Failed (Constraint) Given course has associated classes When Staff attempts to delete Then system shows error as defined in Business Rules AC6 – Data Persistence After Delete
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	Given course is deleted successfully When list reloads Then deleted course no longer appears AC7 – System Error Handling Given system fails to delete course When Staff confirms deletion Then system shows error as defined in Business Rules AC8 – UI Update After Delete Given deletion is successful When system updates UI Then course list refreshes automatically
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5. USER STORY: Staff Create Class

Card	- As a Staff - I want to create a new class - So that classes can be organized under courses for student enrollment
Summary	This user story allows Staff to create a class associated with a course by entering required class information
In-Scope:	- Input class details, schedule info, and assign the class to a specific course.

Out of scope	
Trigger	Staff clicks the [Add New] button on the Class Management screen.
Pre-condition	- Staff is logged in - Staff has permission to manage classes - At least one course exists in the system - System database is available
Business rule and error message	- Class must belong to an existing course → Error: "Selected course does not exist." - Course must be selected → Error: "Please select a course." - Class name must not be empty → Error: "Class name is required." - Class name must be unique within the same course → Error: "Class name already exists for this course." - Teacher name must not be empty → Error: "Teacher name is required." - Capacity must be greater than 0 → Error: "Capacity must be greater than 0." - Start date must be today or future → Error: "Start date must be today or a future date." - <i>Schedule info must not be empty</i>

	Error: "Schedule information is required." - System must successfully save class Error: "Unable to create class."
Screen design	 
Screen description	Create Class screen allows Staff to enter class information.

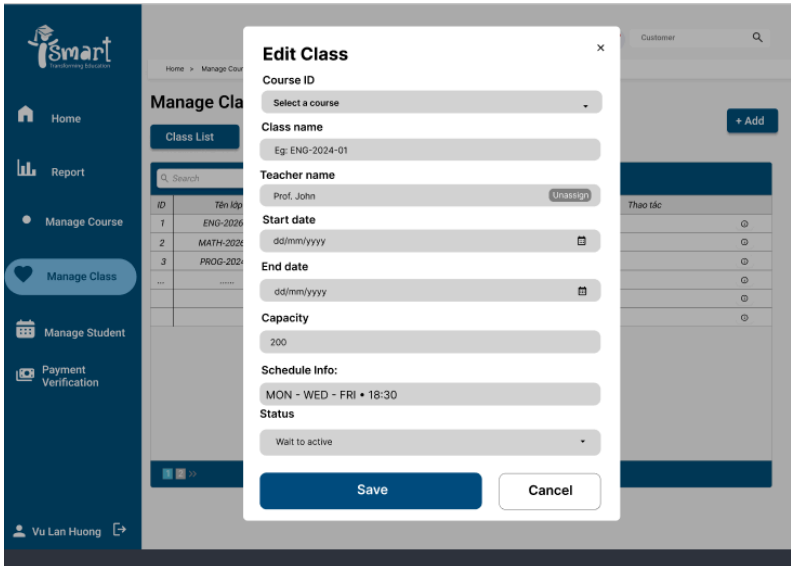
	<table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>course_id</td><td>Course that class belongs to</td><td>int</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr><tr><td>class_name</td><td>Name of class</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>Yes</td></tr><tr><td>teacher_id</td><td>Name of assigned teacher</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>no</td></tr><tr><td>start_date</td><td>Class start date</td><td>date</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr><tr><td>end_date</td><td>Class end date</td><td>date</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr><tr><td>capacity</td><td>Maximum number of students</td><td>number</td><td>1</td><td>200</td><td>-</td><td>Yes</td></tr><tr><td>schedule_info</td><td>Class schedule (e.g., Mon-We d 18:00)</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>yes</td></tr><tr><td>status</td><td>Class status (Active/Inactive)</td><td>varchar</td><td>-</td><td>-</td><td>20</td><td>Yes</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	course_id	Course that class belongs to	int	-	-	-	Yes	class_name	Name of class	varchar	1	100	100	Yes	teacher_id	Name of assigned teacher	varchar	1	100	100	no	start_date	Class start date	date	-	-	-	Yes	end_date	Class end date	date	-	-	-	Yes	capacity	Maximum number of students	number	1	200	-	Yes	schedule_info	Class schedule (e.g., Mon-We d 18:00)	varchar	1	100	100	yes	status	Class status (Active/Inactive)	varchar	-	-	20	Yes
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<table><tr><td rowspan="3">Acceptance Criteria</td><td>AC1 – Open Create Class Screen</td></tr><tr><td>Given Staff is on Class Management page When Staff clicks “Add New” Then system displays Create Class screen</td></tr><tr><td>AC2 – Create Class Successfully</td></tr></table>	Acceptance Criteria	AC1 – Open Create Class Screen	Given Staff is on Class Management page When Staff clicks “Add New” Then system displays Create Class screen	AC2 – Create Class Successfully																																																												
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	Given Staff enters valid class information When Staff clicks Create Then system saves class successfully And redirects to Class List AC3 – Class Name Required Given Staff leaves class_name empty When submitting form Then system shows error as defined in Business Rules AC4 – Duplicate Class Name in Course Given class name already exists in selected course When submitting form Then system shows error as defined in Business Rules AC5 – Invalid Course Selection Given selected course does not exist When submitting form Then system shows error as defined in Business Rules AC6 – Invalid Start Date Given Staff selects a past date When submitting form
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	Then system shows error as defined in Business Rules AC7 – Invalid Capacity Given Staff enters capacity ≤ 0 When submitting form Then system shows error as defined in Business Rules AC8 – Schedule Info Required Given Staff leaves schedule_info empty When submitting form Then system shows error as defined in Business Rules. AC9 – Cancel Create Class Given Staff is on Create Class screen When Staff clicks Cancel Then system returns to Class List And no data is saved AC10 – System Error Handling Given system fails to save class When submitting form Then system shows error as defined in Business Rules
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6. USER STORY: Staff Update Class

Card	- As a Staff - I want to update class information - So that class data remains accurate for course organization
Summary	This user story allows Staff to update existing class information such as class name, assigned course, teacher, start date, and capacity
In-Scope:	- Modify existing class details, schedule, capacity, and change operational status.
Out of scope	- Delete a class
Trigger	Staff clicks the [Edit] icon on a specific class record.
Pre-condition	- Staff is logged in - Class exists in the system - Staff has permission to update class - System database is available
Business rule and error message	- Class must belong to an existing course → Error: "Selected course does not exist." - Course must be selected → Error: "Please select a course." - Class name must not be empty → Error: "Class name is required." - Class name must be unique within the same course (excluding current class)

	→ Error: "Class name already exists for this course." - Teacher name must not be empty → Error: "Teacher name is required." - Capacity must be greater than 0 → Error: "Capacity must be greater than 0." - Start date must be today or future → Error: "Start date must be today or a future date." - <i>Schedule info must not be empty</i> Error: "Schedule information is required." - System must successfully update class → Error: "Unable to update class."							
Screen design								
Screen description	Update Class screen allows Staff to modify existing class information. <table><tr><th>Name</th><th>Descripti on</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlen gth</th><th>Require d</th></tr></table>	Name	Descripti on	Type	Min	Max	Maxlen gth	Require d
Name	Descripti on	Type	Min	Max	Maxlen gth	Require d		

	course_id	Course that class belongs to	int	-	-	-	Yes
	class_name	Name of class	varchar	1	100	100	Yes
	teacher_id	assigned teacher	varchar	1	100	100	no
	start_date	Class start date	date	-	-	-	Yes
	end_date	Class end date	date	-	-	-	Yes
	capacity	Maximum number of students	number	1	200	-	Yes
	schedule_info	Class schedule (e.g., Mon-We d 18:00)	varchar	1	100	100	yes
	status	Class status (Active/Inactive)	varchar	-	-	20	Yes
Acceptance Criteria	AC1 – Open Update Class Screen Given Staff is on Class Management page When Staff clicks Edit on a class Then system displays Update Class screen And shows existing class data AC2 – Update Class Successfully						

Given Staff updates valid class information
When Staff clicks Save
Then system updates class successfully
And redirects to Class List

AC3 – Class Name Required

Given Staff clears class_name
When submitting form
Then system shows error as defined in Business Rules

AC4 – Duplicate Class Name in Course

Given another class in the same course has the same name
When submitting form
Then system shows error as defined in Business Rules

AC5 – Invalid Course Selection

Given selected course does not exist
When submitting form
Then system shows error as defined in Business Rules

AC6 – Invalid Start Date

Given Staff selects a past date
When submitting form

	Then system shows error as defined in Business Rules AC7 – Invalid Capacity Given Staff enters capacity ≤ 0 When submitting form Then system shows error as defined in Business Rules AC8 – Schedule Info Required Given Staff clears schedule_info When submitting form Then system shows error as defined in Business Rules. AC9 – Activate Class / Update Status Given class status is currently "Wait to active" When Staff changes Status to "Active" and clicks Save Then system updates the status successfully and makes the class available for enrollment operations. AC10 – Cancel Update Given Staff is on Update Class screen When Staff clicks Cancel Then system returns to Class List And no changes are saved AC11 – System Error Handling
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	Given system fails to update class When submitting form Then system shows error as defined in Business Rules
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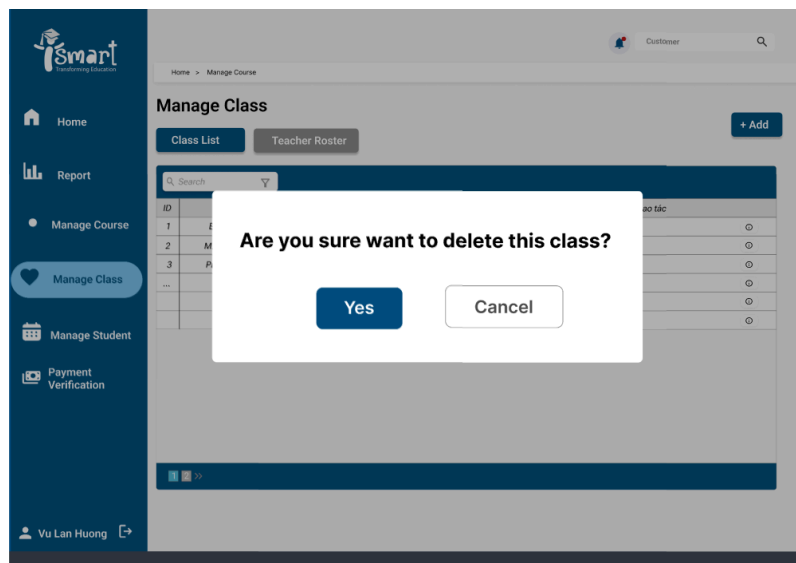
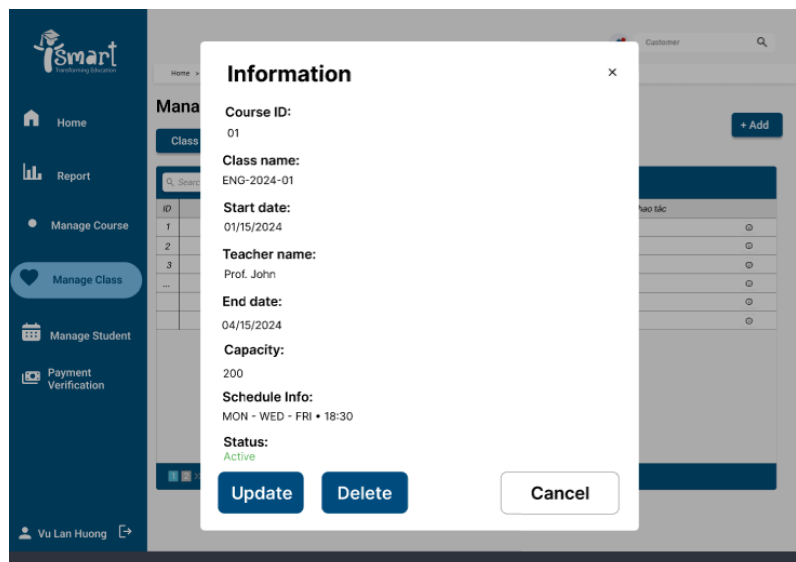
7. USER STORY: Staff Delete Class

Card	- As a Staff - I want to delete a class - So that unnecessary or unused classes are removed from the system
Summary	This user story allows Staff to delete a class with validation to ensure that no students are enrolled in the class
In-Scope:	- Delete a class with strict constraint validation (blocked if students are enrolled).
Out of scope	- Recover deleted classes.
Trigger	Staff clicks Delete button on a class in Class Management screen
Pre-condition	- Staff is logged in - Class exists in the system - Staff has permission to delete class - System database is available

Business rule and error message

- Staff must confirm before deletion
→ **Message:** "Are you sure you want to delete this class?"
- Class cannot be deleted if students are enrolled
→ **Error:** "Class cannot be deleted because students are enrolled."
- System must successfully delete class
→ **Error:** "Unable to delete class."

Screen design



Screen description	Class Management screen allows Staff to delete classes. Includes: • Class table • Delete button for each class • Confirmation dialog: ○ Confirm button ○ Cancel button																																																															
	<table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>course_id</td><td>Course that class belongs to</td><td>int</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr><tr><td>class_name</td><td>Name of class</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>Yes</td></tr><tr><td>teacher_id</td><td>assigned teacher</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>no</td></tr><tr><td>start_date</td><td>Class start date</td><td>date</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr><tr><td>end_date</td><td>Class end date</td><td>date</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr><tr><td>capacity</td><td>Maximum number of students</td><td>number</td><td>1</td><td>200</td><td>-</td><td>Yes</td></tr><tr><td>schedule_info</td><td>Class schedule (e.g., Mon-Wed 18:00)</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>yes</td></tr><tr><td>status</td><td>Class status (Active/Inactive)</td><td>varchar</td><td>-</td><td>-</td><td>20</td><td>Yes</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	course_id	Course that class belongs to	int	-	-	-	Yes	class_name	Name of class	varchar	1	100	100	Yes	teacher_id	assigned teacher	varchar	1	100	100	no	start_date	Class start date	date	-	-	-	Yes	end_date	Class end date	date	-	-	-	Yes	capacity	Maximum number of students	number	1	200	-	Yes	schedule_info	Class schedule (e.g., Mon-Wed 18:00)	varchar	1	100	100	yes	status	Class status (Active/Inactive)	varchar	-	-	20	Yes
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status	Class status (Active/Inactive)	varchar	-	-	20	Yes																																																										
Acceptance Criteria	AC1 – Trigger Delete Action																																																															

	Given Staff is on Class Management page When Staff clicks Delete on a class Then system displays confirmation dialog AC2 – Confirm Deletion Given confirmation dialog is displayed When Staff clicks Confirm Then system proceeds with deletion AC3 – Cancel Deletion Given confirmation dialog is displayed When Staff clicks Cancel Then system closes dialog And no data is deleted AC4 – Delete Class Successfully Given class has no students enrolled When Staff confirms deletion Then system deletes class And removes it from the list AC5 – Delete Class Failed (Constraint) Given class has students enrolled When Staff attempts to delete Then system shows error as defined in Business Rules AC6 – Data Persistence After Delete Given class is deleted successfully When list reloads Then deleted class no longer appears AC7 – UI Refresh After Delete
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	Given deletion is successful When system updates UI Then class list refreshes automatically AC8 – System Error Handling Given system fails to delete class When Staff confirms deletion Then system shows error as defined in Business Rules AC9 – Delete Non-existing Class Given class does not exist in system When Staff attempts to delete Then system shows error as defined in Business Rules
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8. USER STORY: Staff View Class

Card	- As a Staff - I want to view and filter classes - So that I can access class information for operations or consultation
Summary	This user story allows users to view class list, search classes, filter by course, and view detailed class information
In-Scope:	- View class list, view detailed information, search by name, and filter by course.

Out of scope	- Modify class data.
Trigger	Staff clicks the [Manage Class] menu on the Sidebar.
Pre-condition	- User is logged in - Classes exist in the system - System database is available
Business rule and error message	- System must retrieve class data from database → Error: "Unable to load class data." - Search keyword must be applied to class name → Behavior: return matching results - Filter must be based on valid course → Error: "Invalid course filter." - If no class matches search/filter → Message: "No classes found." - Only authorized users can access class list → Error: "You do not have permission to perform this action." - Session must be valid → Error: "Session expired. Please log in again."
Screen design	

	 Home Report Manage Course Manage Class Manage Student Payment Verification Vu Lan Huong Home > Manage Course Manage Class Class List Teacher Roster + Add Q Search <table> <tr> <th>ID</th><th>Tên lớp</th><th>Khóa học</th><th>Trạng thái</th><th>Thao tác</th></tr> <tr> <td>1</td><td>ENG-2026-01</td><td>English for Beginners</td><td>Active</td><td></td></tr> <tr> <td>2</td><td>MATH-2026-01</td><td>Advanced Mathematics</td><td>Active</td><td></td></tr> <tr> <td>3</td><td>PROG-2024-01</td><td>Programming Fundamentals</td><td>Wait for active</td><td></td></tr> <tr> <td>...</td><td>.....</td><td>.....</td><td>Finish</td><td></td></tr> <tr> <td></td><td></td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td></td><td></td></tr> </table> Information Course ID: 01 Class name: ENG-2024-01 Start date: 01/15/2024 Teacher name: Prof. John End date: 04/15/2024 Capacity: 200 Schedule Info: MON - WED - FRI • 18:30 Status: Active Update Delete Cancel	ID	Tên lớp	Khóa học	Trạng thái	Thao tác	1	ENG-2026-01	English for Beginners	Active		2	MATH-2026-01	Advanced Mathematics	Active		3	PROG-2024-01	Programming Fundamentals	Wait for active		...			Finish											
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...			Finish																																	
																																				
																																				
Screen description	Class Management screen allows Staff to view and search class information. <table> <tr> <th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>MaxLength</th><th>Required</th></tr> <tr> <td>course_id</td><td>Course that class belongs to</td><td>int</td><td>-</td><td>-</td><td>-</td><td>Yes</td></tr> <tr> <td>class_name</td><td>Name of class</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>Yes</td></tr> <tr> <td>teacher_id</td><td>assigned teacher</td><td>varchar</td><td>1</td><td>100</td><td>100</td><td>no</td></tr> </table>	Name	Description	Type	Min	Max	MaxLength	Required	course_id	Course that class belongs to	int	-	-	-	Yes	class_name	Name of class	varchar	1	100	100	Yes	teacher_id	assigned teacher	varchar	1	100	100	no							
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	Given Staff is on class list When Staff selects a class Then system displays detailed class information AC4 – Search Class Given Staff enters keyword When performing search Then system displays matching classes AC5 – Filter Class by Course Given Staff selects a course in filter When filter is applied Then system displays classes belonging to that course AC6 – No Class Available Given no classes exist When page loads Then system shows message as defined in Business Rules AC7 – No Search/Filter Result Given no classes match search/filter When action is performed Then system shows message as defined in Business Rules
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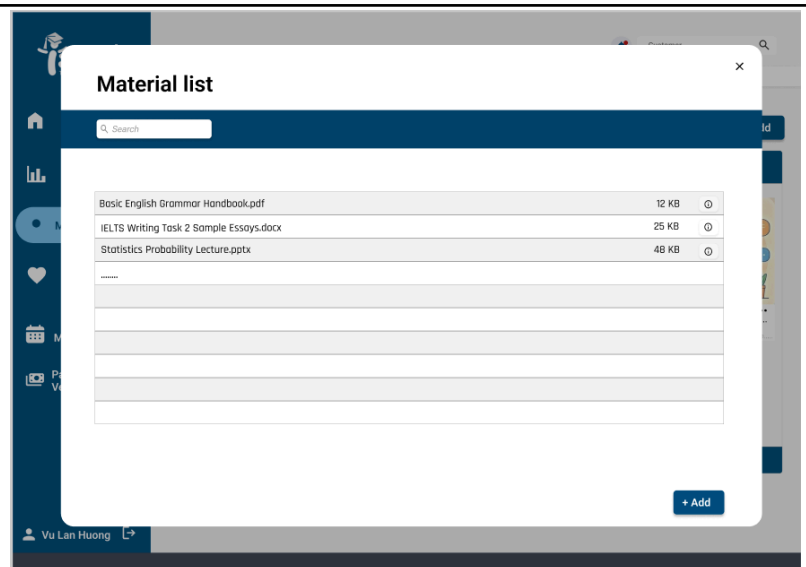
	AC8 – Navigate to Create Class Given Staff clicks “Add” Then system navigates to Create Class screen AC9 – System Error Handling Given system fails to load class data When page loads Then system shows error as defined in Business Rules
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9. USER STORY: Staff View Material

Card	- As a Staff - I want to view the list of materials - So that I can access learning resources for consultation and support purposes
Summary	This user story allows Users (Staff and Sale) to view materials in the system, including searching, filtering, and viewing material details by course
In-Scope:	- View and search materials strictly associated with a specific course.

Out of scope	
Trigger	Staff clicks the specific course and [Add] button Materail
Pre-condition	- User is logged into the system - Materials may or may not exist - System database is available
Business rule and error message	- User role is only allowed to view material information → Error: "You do not have permission to modify material information." - System must retrieve material data from database → Error: "Unable to load material data." - If no materials exist → Message: "No materials found." - If search/filter returns no result → Message: "No materials found." - Search keyword must match material title or course name → System returns matching results - Course filter must reference existing course → Error: "Invalid course selection." - Only active session users can access this screen → Error: "Session expired. Please log in again."

Screen design



Screen description

Material List screen allows User to view and search materials.

Includes:

- Search bar (by material title)
- Course filter dropdown
- Material table
- Clickable row / View button → open material detail

No Add / Edit / Delete buttons

Name	Description	Type	Min	Max	Maxlength	Required
title	Material title	varchar	1	255	255	yes
file_url	File link or material URL	varchar	1	500	500	yes
course_id	Course of the material	int	-	-	-	yes
created_by	Creator	int	-	-	-	yes
created_date	Material creation date	date time	-	-	-	yes

	<table><tr><td>keyword</td><td>Search keyword</td><td>varchar</td><td>0</td><td>100</td><td>100</td><td>no</td></tr><tr><td>filter_course</td><td>Selected course for filtering</td><td>int</td><td>-</td><td>-</td><td>-</td><td>no</td></tr></table>	keyword	Search keyword	varchar	0	100	100	no	filter_course	Selected course for filtering	int	-	-	-	no
keyword	Search keyword	varchar	0	100	100	no									
filter_course	Selected course for filtering	int	-	-	-	no									
Acceptance Criteria	AC1 – View Material List Given Staff opens Material List page When the system loads Then the system displays the list of materials AC2 – Display Material Information Given material list is displayed Then each material shows title, course AC3 – View Material Detail Given Staff selects a material When clicking View or row Then system displays material detail AC4 – Search Material Given Staff enters a keyword When performing search Then system displays matching materials AC5 – Filter by Course Given Staff selects a course When filter is applied Then system displays materials belonging to that course AC6 – No Material Found Given no materials match search/filter When system returns result Then system shows message as defined in Business Rules														

	AC7 – Data Load Failure Given system fails to retrieve material data When loading page Then system shows error as defined in Business Rules AC8 – Session Expired Given Staff session has expired When accessing the page Then system redirects to login page And shows message as defined in Business Rules
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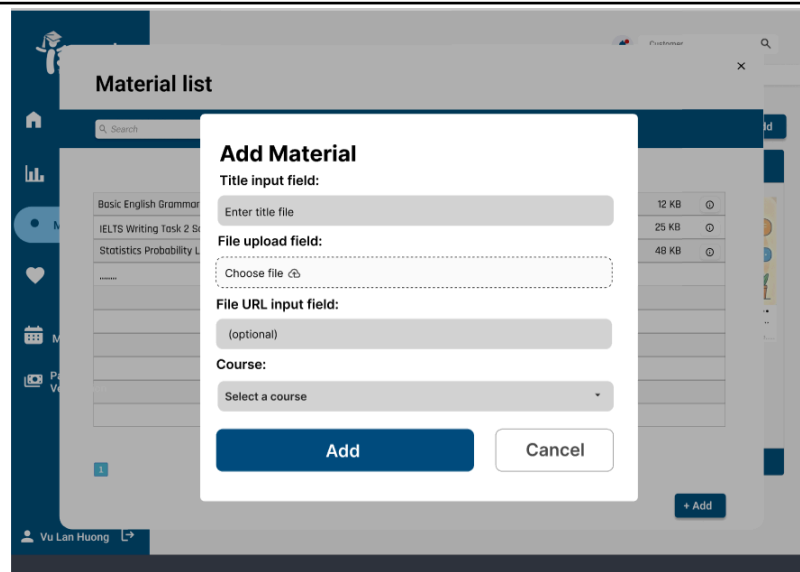
10. USER STORY: Staff Create Material

Card	- As an Staff - I want to create a new material - So that learning resources can be added to courses and used by all classes
Summary	This user story allows Staff to create a new material by entering required information and assigning it to a course
In-Scope:	- Upload file or input URL, auto-extract metadata (format/size), and link to a course.
Out of scope	

Trigger	Staff clicks the specific course and [Add] button Materail.
Pre-condition	- Staff is logged into the system - At least one course exists in the system - System database is available
Business rule and error message	- Material title must not be empty → Error: "Material title is required." - Material title length must be within allowed range (1–255) → Error: "Material title must be between 1 and 255 characters." - At least one source must be provided (file OR file_url) → Error: "Material file or URL is required." - If both file and file_url are provided → System prioritizes file upload (or accept both based on implementation – chọn 1 và giữ consistent) - File must be in supported formats (PDF, DOC, DOCX, PPT, PPTX) → Error: "Unsupported file format." - File size must not exceed system limit (e.g., 20MB) → Error: "File size exceeds allowed limit." - Course must be selected and must exist in the system → Error: "Selected course does not exist."

- System must successfully save material
→ Error: **"Unable to create material."**

Screen design



Screen description

Create Material screen allows Staff to add new learning materials.

Includes:

- Title input field
- File upload field
- File URL input field
- Course dropdown (assign material to course)
- Add button
- Cancel button

Admin must provide:

- File OR URL (at least one)

Name	Description	Type	Min	Max	Maxlength	Required
title	Name of the material	varchar	1	255		yes
file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	yes

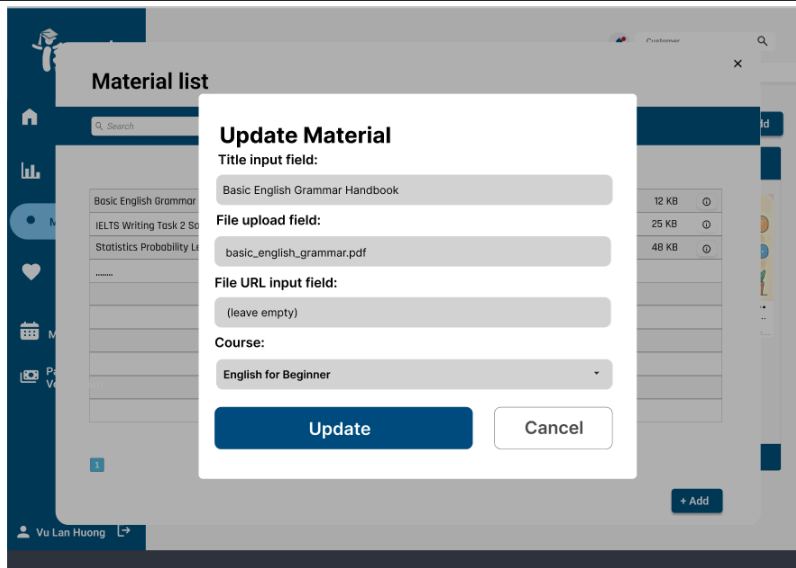
	<table><tr><td>file_size</td><td>Size of the file</td><td>varchar</td><td>1</td><td>20</td><td>-</td><td>yes</td></tr><tr><td>file_url</td><td>Link to access file</td><td>varchar</td><td>-</td><td>-</td><td></td><td>yes</td></tr><tr><td>keyword</td><td>Search title</td><td>varchar</td><td>0</td><td>100</td><td>100</td><td>no</td></tr><tr><td>created_at</td><td>Creation date</td><td>date time</td><td>-</td><td>-</td><td>-</td><td>yes</td></tr></table>	file_size	Size of the file	varchar	1	20	-	yes	file_url	Link to access file	varchar	-	-		yes	keyword	Search title	varchar	0	100	100	no	created_at	Creation date	date time	-	-	-	yes
file_size	Size of the file	varchar	1	20	-	yes																							
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keyword	Search title	varchar	0	100	100	no																							
created_at	Creation date	date time	-	-	-	yes																							
Acceptance Criteria	AC1 – Open Create Material Screen Given Staff is on Material Management page When Staff clicks Add New Material Then system displays Create Material screen AC2 – Create Material Successfully Given Staff enters valid material information When Staff clicks Add Then system saves the material And redirects to Material List And displays the new material in the list AC3 – Title Required Given Staff leaves title empty When submitting the form Then system shows error as defined in Business Rules AC4 – Title Length Validation Given title length is less than 1 or greater than 255 When submitting the form Then system shows error as defined in Business Rules AC5 – Missing File and URL Given both file and file_url are empty When submitting the form Then system shows error as defined in Business Rules																												

	AC6 – Invalid Course Selection Given selected course does not exist When submitting the form Then system shows error as defined in Business Rules AC7 – Unsupported File Format Given Staff uploads an unsupported file When submitting the form Then system shows error as defined in Business Rules AC8 – Cancel Create Given Staff is on Create Material screen When Staff clicks Cancel Then system returns to Material List And no data is saved AC9 – System Error Handling Given system fails to save material When Staff submits the form Then system shows error as defined in Business Rules
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11. USER STORY: Staff Update Material

Card	- As an Staff - I want to update material information - So that learning resources remain accurate and up to date
Summary	This user story allows Staff to modify existing material information including title, file (or URL), and assigned course

In-Scope:	- Replace file/URL, update title, and auto-re-extract metadata.
Out of scope	- Delete material
Trigger	Staff clicks the [Edit] icon on a specific material record.
Pre-condition	- Staff is logged into the system - Material exists in the system - System database is available
Business rule and error message	- Material title must not be empty → Error: "Material title is required." - Material title length must be within allowed range (1–255) → Error: "Material title must be between 1 and 255 characters." - At least one source must exist (existing file OR new file OR file_url) → Error: "Material file or URL is required." - If both new file and file_url are provided → System prioritizes file upload (hoặc define rõ rule, miễn consistent) - File must be in supported formats (PDF, DOC, DOCX, PPT, PPTX) → Error: "Unsupported file format." - File size must not exceed system limit (e.g., 20MB) → Error: "File size exceeds allowed limit."

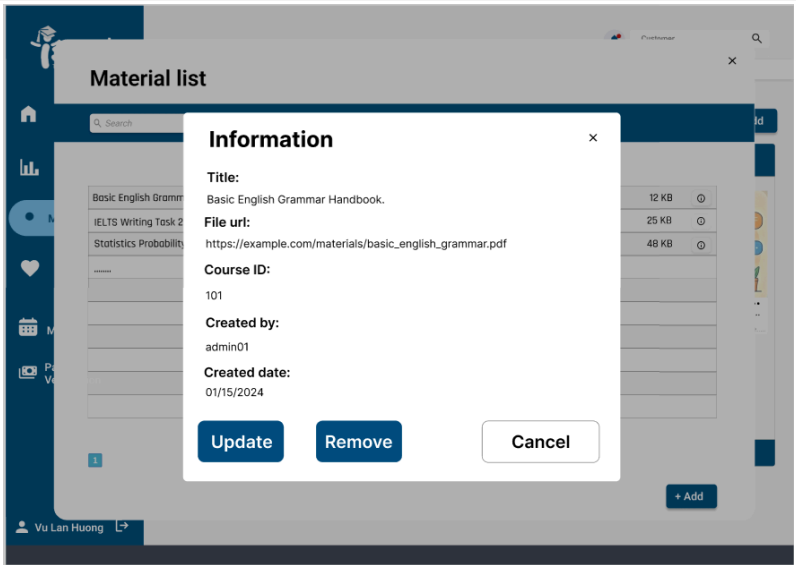
	- Course must be selected and must exist → Error: "Selected course does not exist." - System must successfully update material → Error: "Unable to update material."																																																	
Screen design																																																		
Screen description	Update Material screen allows Staff to edit material information. <table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>title</td><td>Name of the material</td><td>varchar</td><td>1</td><td>255</td><td></td><td>yes</td></tr><tr><td>file_format</td><td>Format (PDF, PPTX, etc.)</td><td>varchar</td><td>1</td><td>10</td><td>10</td><td>yes</td></tr><tr><td>file_size</td><td>Size of the file</td><td>varchar</td><td>1</td><td>20</td><td>-</td><td>yes</td></tr><tr><td>file_url</td><td>Link to access file</td><td>varchar</td><td>-</td><td>-</td><td></td><td>yes</td></tr><tr><td>keyword</td><td>Search title</td><td>varchar</td><td>0</td><td>100</td><td>100</td><td>no</td></tr><tr><td>created_at</td><td>Creation date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>yes</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	title	Name of the material	varchar	1	255		yes	file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	yes	file_size	Size of the file	varchar	1	20	-	yes	file_url	Link to access file	varchar	-	-		yes	keyword	Search title	varchar	0	100	100	no	created_at	Creation date	datetime	-	-	-	yes
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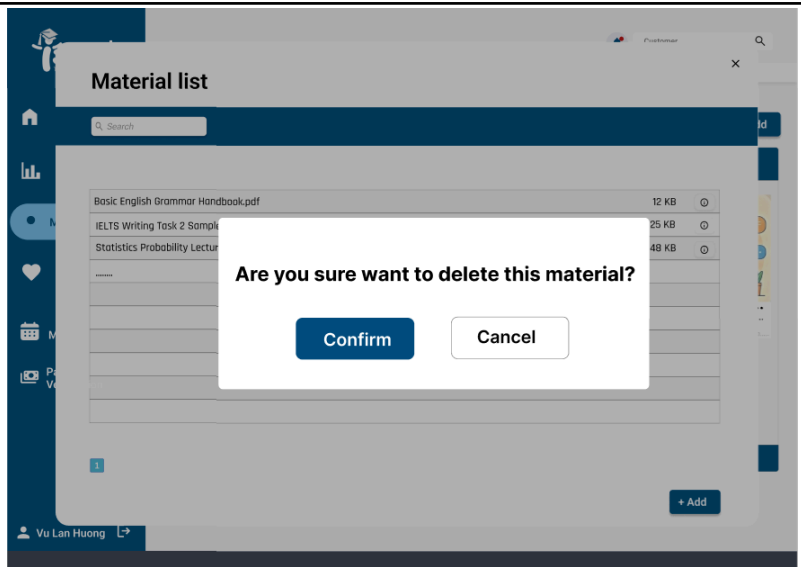
Acceptance Criteria	AC1 – Open Update Material Screen Given Staff is on Material Management page When Staff clicks Edit on a material Then system displays Update Material screen And shows existing material information AC2 – Update Material Successfully Given Staff updates valid information When Staff clicks Save Then system updates the material successfully And redirects to Material List AC3 – Title Required Given Staff clears title When submitting the form Then system shows error as defined in Business Rules AC4 – Title Length Validation Given title is less than 1 or greater than 255 characters When submitting the form Then system shows error as defined in Business Rules AC5 – Missing File Source Given no existing file and no new file or URL provided When submitting the form Then system shows error as defined in Business Rules AC6 – Invalid Course Selection Given selected course does not exist When submitting the form

	Then system shows error as defined in Business Rules AC7 – Unsupported File Format Given Staff uploads unsupported file When submitting the form Then system shows error as defined in Business Rules AC8 – Cancel Update Given Staff is on Update Material screen When Staff clicks Cancel Then system returns to Material List And no changes are saved AC9 – System Error Handling Given system fails to update material When Staff submits the form Then system shows error as defined in Business Rules
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12. USER STORY: Staff Delete Material

Card	- As an Staff - I want to delete a material - So that outdated or incorrect learning resources can be removed from the system
Summary	This user story allows Staff to delete an existing material from the system while ensuring data consistency
In-Scope:	- Permanently remove a material from a course.

Out of scope	- Recover deleted materials.
Trigger	Staff clicks the [Delete] icon on a specific material record.
Pre-condition	- Staff is logged into the system - Material exists in the system - System database is available
Business rule and error message	- Material must exist in the system → Error: "Material does not exist." - System must confirm before deletion → Message: "Are you sure you want to delete this material?" - Material deletion must be permanent (hard delete) - System must successfully delete material → Error: "Unable to delete material."
Screen design	 The screenshot shows a web application interface for managing materials. A modal dialog titled 'Information' is open, displaying details for a selected material. The background shows a table of materials with columns for title, file size, and actions. The modal dialog contains the following information: Title: Basic English Grammar Handbook. File url: https://example.com/materials/basic_english_grammar.pdf Course ID: 101 Created by: admin01 Created date: 01/15/2024 At the bottom of the modal, there are three buttons: 'Update' (blue), 'Remove' (blue), and 'Cancel' (white). The background interface includes a sidebar with navigation icons and a top bar with a search bar and a user profile.

																																																		
Screen description	Material Management screen includes: <table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>title</td><td>Name of the material</td><td>varchar</td><td>1</td><td>255</td><td></td><td>yes</td></tr><tr><td>file_format</td><td>Format (PDF, PPTX, etc.)</td><td>varchar</td><td>1</td><td>10</td><td>10</td><td>yes</td></tr><tr><td>file_size</td><td>Size of the file</td><td>varchar</td><td>1</td><td>20</td><td>-</td><td>yes</td></tr><tr><td>file_url</td><td>Link to access file</td><td>varchar</td><td>-</td><td>-</td><td></td><td>yes</td></tr><tr><td>keyword</td><td>Search title</td><td>varchar</td><td>0</td><td>100</td><td>100</td><td>no</td></tr><tr><td>created_at</td><td>Creation date</td><td>datetime</td><td>-</td><td>-</td><td>-</td><td>yes</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	title	Name of the material	varchar	1	255		yes	file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	yes	file_size	Size of the file	varchar	1	20	-	yes	file_url	Link to access file	varchar	-	-		yes	keyword	Search title	varchar	0	100	100	no	created_at	Creation date	datetime	-	-	-	yes
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keyword	Search title	varchar	0	100	100	no																																												
created_at	Creation date	datetime	-	-	-	yes																																												
Acceptance Criteria	AC1 – Display Delete Option Given material list is displayed Then each material has a Delete button AC2 – Trigger Delete Action Given Staff clicks Delete on a material When action is triggered Then system displays confirmation dialog																																																	

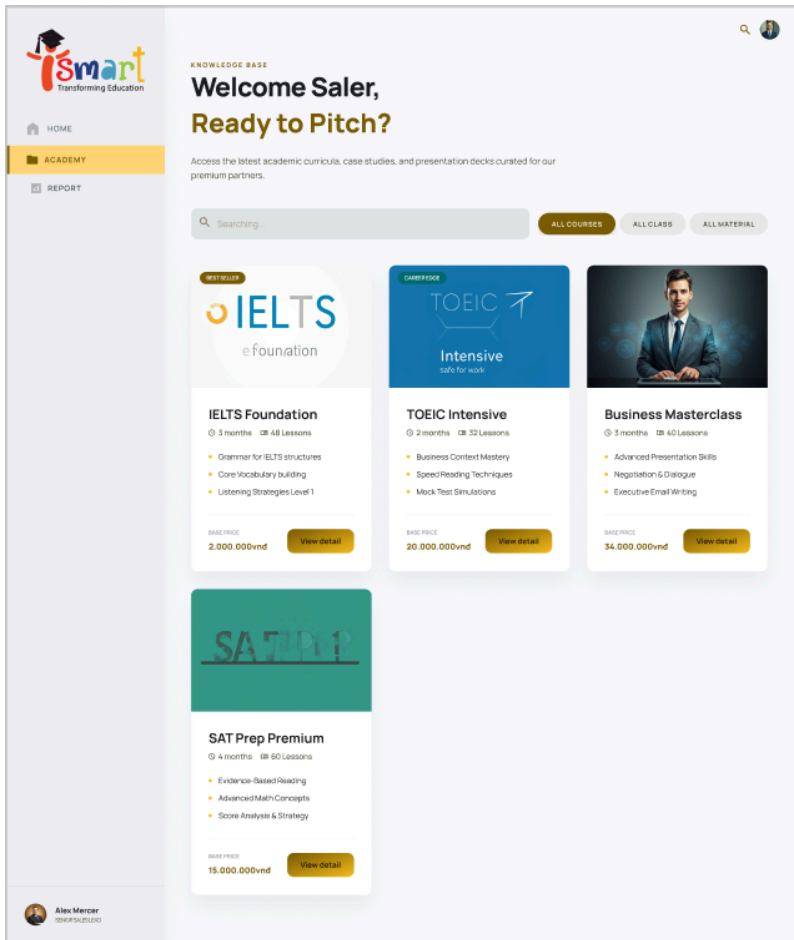
	AC3 – Confirm Delete Successfully Given Staff confirms deletion When system processes request Then material is removed from the system And no longer appears in the list AC4 – Cancel Delete Given confirmation dialog is displayed When Staff clicks Cancel Then system closes dialog And no data is deleted AC5 – Material Not Found Given material does not exist When Staff attempts to delete Then system shows error as defined in Business Rules AC6 – System Error Handling Given system fails to delete material When Staff confirms deletion Then system shows error as defined in Business Rules AC7 – Refresh Material List After Delete Given material is deleted successfully When system updates UI Then material list is refreshed And deleted item is not displayed AC8 – Multiple Delete Attempts Given material has already been deleted When Staff attempts to delete again Then system shows error as defined in Business Rules
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	AC9 – Delete Without Affecting Other Data Given material is deleted When system processes deletion Then no other materials or course data are affected
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Sale

1. USER STORY: Sale View Course

Card	- As a Sale - I want to view and search courses in a catalog format - So that I can quickly access course information and open the Course Detail Hub for customer consultation
Summary	This user story allows Sale users to view the list of courses in a visual Card layout, search for specific courses, and click to access the centralized Course Detail Hub.
In-Scope:	- View the course catalog in Card layout, search by keyword, and click to open specific Course Details.
Out of scope	- Create, update, or delete course data.
Trigger	Sale clicks the [Academy] menu on the left Sidebar (defaults to the "ALL COURSES" tab).
Pre-condition	- Sale is logged in - Courses exist in the system - System database is available
Business rule and error message	- System must retrieve course data from database → Error: "Unable to load course data."

	- Search keyword must be applied to course name → Behavior: return matching results - If no course matches search result → Message: "No courses found." - Session must be valid → Error: "Session expired. Please log in again."
Screen design	 The screenshot displays the 'iSmart' Knowledge Base interface. On the left is a sidebar with navigation links: HOME, ACADEMY (highlighted), and REPORT. The main content area features a 'Welcome Saler, Ready to Pitch?' message and a search bar. Below the search bar, there are four course cards in a grid layout: IELTS Foundation: 3 months, 48 Lessons. Features include Grammar for IELTS structures, Core Vocabulary building, and Listening Strategies Level 1. Base price: 2,000,000vnd. TOEIC Intensive: 2 months, 32 Lessons. Features include Business Context Mastery, Speed Reading Techniques, and Mock Test Simulations. Base price: 20,000,000vnd. Business Masterclass: 3 months, 40 Lessons. Features include Advanced Presentation Skills, Negotiation & Dialogue, and Executive Email Writing. Base price: 34,000,000vnd. SAT Prep Premium: 4 months, 60 Lessons. Features include Evidence-Based Reading, Advanced Math Concepts, and Score Analysis & Strategy. Base price: 15,000,000vnd. Each card has a 'View detail' button. The bottom left corner shows a user profile for Alex Mercer.
Screen description	Search input field & Search button Course Card View (Grid layout) Includes: ● Course Name

- Duration
- Fee
- Status
- Course detail view (popup or separate page)

Name	Description	Type	Min	Max	Maxlength	Required
course_name	Name of the course	varchar	1	100	100	yes
description	Description of the course	text	-	-	500	no
duration	Course duration	number	1	999	-	yes
fee	Course tuition fee	Decimal	1	-	-	yes
keyword	Search keyword	varchar	0	100	100	no
status	Course status	varchar	-	-	-	yes

Acceptance Criteria

AC1 – View Course Catalog

Given Sale opens Course page

When system loads

Then system displays courses in a visual Card format

AC2 – Display Course Card Information

Given course catalog is displayed

When viewing the cards

Then each course card shows name, duration, fee, and status

AC3 – Navigate to Course Detail Hub

Given course catalog is displayed

When Sale clicks a Course Card or Course

Then system navigates to the Course Detail Hub
(displaying related classes and materials)

AC4 – Search Course

Given Sale enters keyword

When Sale clicks Search

Then system displays matching course cards

AC5 – Search Result Accuracy

Given keyword matches course name

When search is performed

Then system returns correct results

AC6 – No Course Found

Given no course matches keyword

When search is performed

Then system shows message as defined in Business
Rules

	AC7 – Empty Course Catalog Given no courses exist in system When Sale opens Course page Then system shows message as defined in Business Rules AC8 – System Error Handling Given system fails to retrieve data When loading course catalog Then system shows error as defined in Business Rules
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2. USER STORY: Sale View Classes

Card	- As a Sale - I want to view classes in "Wait to active" or "Open" status within the Course Detail Hub - So that I can identify which classes need more students and consult customers to fill those slots
Summary	This user story allows Sale users to focus on upcoming and recruiting classes. It provides

	visibility into the "Wait to active" classes where enrollment is still possible, helping Sale drive revenue.
In-Scope:	- View global or course-specific class lists, check real-time capacity, and identify "Wait to active"/"Open" classes.
Out of scope	- Modify class data or operational statuses.
Trigger	Sale clicks the [ALL CLASS] toggle on the Academy screen, OR views the Classes section inside a Course Detail Hub.
Pre-condition	- Sale is logged in - System database is available - The selected course exists and may have associated classes
Business rule and error message	- System must retrieve classes of the selected course, prioritizing those NOT YET started. Display clear status labels: "Wait to active" (Chờ khai giảng), "Recruiting" (Đang tuyển), or "Full"
Screen design	



 HOME
  ACADEMY
  REPORT

KNOWLEDGE BASE

Welcome Saler,

Ready to Pitch?

Access the latest academic curricula, case studies, and presentation decks curated for our premium partners.

IELTS PREP



Live IELTS 01

📅 Mon-Wed-Fri 18:00

AVAILABILITY

12/15 students LIMITED SEATS





TEACHER
Dr. Simon K.



CORPORAL ESL

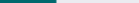


Beginner ESL 04

📅 Tue-Thu 19:30

AVAILABILITY

8/20 students OPEN





TEACHER
Sarah Miller



BUSINESS



Corporate Comm A

📅 Sat 09:00 - 12:00

AVAILABILITY

14/15 students 1 SEAT LEFT





TEACHER
Prof. Albert G.



IELTS PREP



Mastering IELTS 02

📅 Mon-Fri 20:05

AVAILABILITY

5/15 students OPEN





TEACHER
Emily Chen





Alex Mercer
@ALEXMERCE123

Screen description

Wait to active/ Full/ Active

Capacity Tracking

Course Card View (Grid layout)

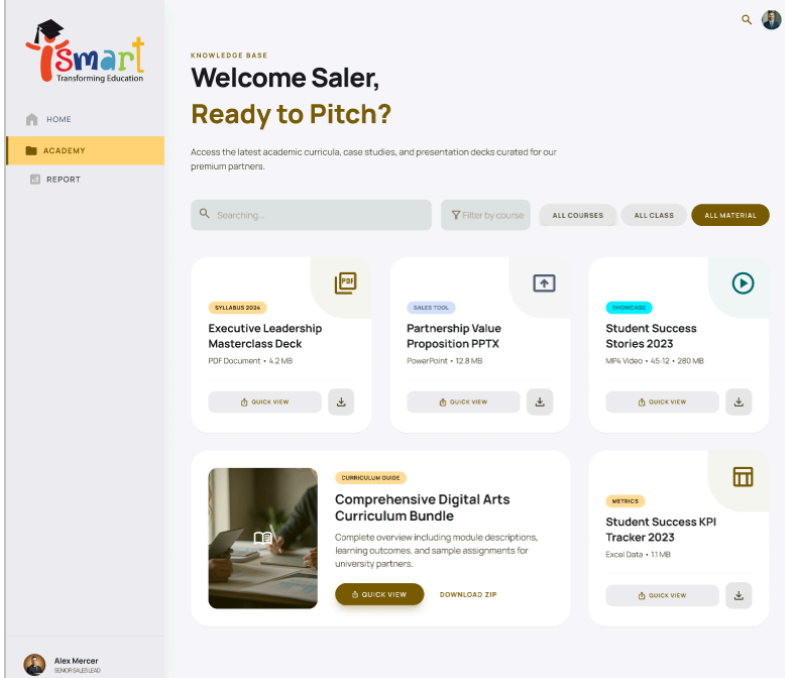
Name	Description	Type	Min	Max	Maxlength	Required
class_name	Name of the class	varchar	1	100	100	yes
schedule info	Time/Days	text	1	100	100	yes
teacher_name	Name of teacher	varchar	1	100	100	Yes
start_date	Class start date	date	-	-	-	Yes
end_date	Class end date	date	-	-	-	Yes
available_slots	max capacity - current_enrollment	number				yes
status	Course status	varchar	-	-	-	yes

Acceptance Criteria	AC1 – Display Upcoming Classes Given Sale is in the Course Detail Hub When the class list loads Then the system must show classes that are "Wait to active" or "Open" at the top. AC2 – Urgent Capacity Display Given Sale views classes, Then the system displays the number of Available Slots, Teacher Name, Schedule, and Start/End dates clearly AC3 – Identify Full Classes Given a class has reached its max capacity Then its status must change to "Full" And Sale should see it as "Not available for enrollment" to avoid consulting wrong info. AC4 – Historical/Active Reference Given a class is already "Open" Then Sale can still view its info (teacher, schedule) but it is marked as "Open". AC5 – Quick Enrollment Given class status is Wait to active or Recruiting, When the card is displayed,

	Then system provides an [Enroll] button to quickly trigger the Enrollment Form.
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3. USER STORY: Sale View Materials

Card	- As a Sale - I want to view learning materials directly within the Course Detail Hub - So that I can quickly showcase or share syllabus and brochures to convince potential customers
Summary	This user story allows Sale users to view, search, and access materials (files/links) specifically associated with the course they are consulting. It prioritizes quick access and visual file information.
In-Scope:	- View global or course-specific learning materials, search by title, and access quick-share links/files.
Out of scope	- Upload, update, or delete materials.
Trigger	Sale clicks the [ALL MATERIAL] toggle on the Academy screen, OR views the Materials section inside a Course Detail Hub.
Pre-condition	- Sale is logged in - The selected course exists and has materials assigned in the database - System database is available

Business rule and error message	- System must retrieve materials ONLY belonging to the current course_id → Error: "Unable to load materials for this course." - If no materials are found for the course Error: "No materials available for this course yet." - If search keyword returns no match → Error: "No materials match your search."																																			
Screen design																																				
Screen description	Materials Section (Grid/Card layout within Course Detail Hub). <table><tr><th>Name</th><th>Description</th><th>Type</th><th>Min</th><th>Max</th><th>Maxlength</th><th>Required</th></tr><tr><td>title</td><td>Name of the material</td><td>varchar</td><td>1</td><td>255</td><td></td><td>yes</td></tr><tr><td>file_format</td><td>Format (PDF, PPTX, etc.)</td><td>varchar</td><td>1</td><td>10</td><td>10</td><td>yes</td></tr><tr><td>file_size</td><td>Size of the file</td><td>varchar</td><td>1</td><td>20</td><td>-</td><td>yes</td></tr><tr><td>file_url</td><td>Link to access file</td><td>varchar</td><td>-</td><td>-</td><td></td><td>yes</td></tr></table>	Name	Description	Type	Min	Max	Maxlength	Required	title	Name of the material	varchar	1	255		yes	file_format	Format (PDF, PPTX, etc.)	varchar	1	10	10	yes	file_size	Size of the file	varchar	1	20	-	yes	file_url	Link to access file	varchar	-	-		yes
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keyword	Search title	varchar	0	100	100	no		
Acceptance Criteria	AC1 – Display Contextual Materials Given Sale is viewing a Course Detail Hub When the Material section loads Then the system displays ONLY materials linked to that specific course AC2 – Visual File Identification Given materials are displayed Then each item must show a specific icon corresponding to its file_format (PDF, Video, etc.) AC3 – Display File Details Given a material card is visible Then Sale can see the title, file format AC4 – Search within Course Given Sale enters a keyword in the material search bar When searching Then system filters materials by title within that course ONLY AC5 – Quick Access/Sharing Given a material is displayed When Sale clicks [Open Link] or [Download]							

	Then the system provides the file/link for sharing with customers AC6 – Permission Guardrail Given Sale is viewing materials Then no options to "Upload", "Edit title", or "Delete" are visible AC7 – Empty State Handling Given the course has no materials assigned When the page loads Then system shows: "No materials available for this course yet."
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